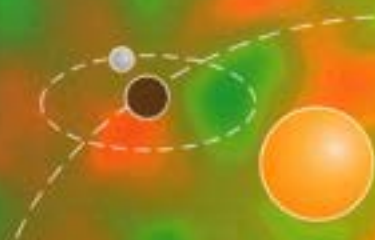


Legends of SCIENCE

BOOK 2

SCIENCE



HOWARD CLARK



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LEGENDS of SCIENCE

Book 2

Howard Clark



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Preface

Legends of Science presents sets of guided questions and activities on a number of diverse topics suitable for Year 9 and 10 students. It is designed to inspire students to extend their understanding, ideas and vocabulary beyond the classroom lessons.

The topics include the life and work of many famous (and some not so famous) scientists — from Aristotle to Wong Staal. The people featured range from the earliest scientific thinkers to current researchers. The variety of scientists included means that the major scientific disciplines of biology, chemistry, geology and physics are covered.

Related extension pages contain further activities in the areas where our scientific legends excelled. The questions range from concrete recall, to comprehension, extraction, analysis and synthesis. These will extend the learning capabilities of all students. Opportunities are provided for students to give their own opinions on scientific issues, and will encourage interest in the more formal activities.

The *Legends of Science* worksheets have a number of uses:

- exercises for when the regular teacher is absent
- pre-testing or post-testing when particular topics are covered
- topic review and revision prior to examinations
- exercises to motivate the disinterested student
- helps Year 10 students choose their Senior Science subjects for matriculation.
- individual extension/self-study exercises for the self-motivated.

A number of icons have been used to provide easy access to the activity involved.

GROUP WORK



Group discussions to encourage communication.

THINK ABOUT IT



Pre-knowledge questions to incorporate earlier learning.

LOOK IT UP



Research to encourage extension activities.

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Nick Jacques (1949–)

Nick Jacques was born in Walsall, England in 1949, and travelled to Australia in his youth. He studied both chemistry and biochemistry at the University of Sydney, graduating with an Honours BSc in 1972 and a PhD in 1977. Jacques' thesis was on the relationship between bacterial surface structure and function. This led him to research into oral microbiology. Jacques continues this research today in his positions as Principal Research Scientist at the Institute of Dental Research, and Head of Oral Biology at the University of Sydney.

- 1 (a) What is chemistry?
(b) What is biochemistry?
- 2 Nick's thesis was on 'The Relationship Between Bacterial Surface Structure and Function'.
(a) What do the words structure and function mean?
(b) What do you think the thesis is about?
- 3 Explain what might be studied in oral microbiology.

The story so far



Since the first humans started to eat, their mouths have contained bacteria. These bacteria come mainly from our mothers and close family members. The bacteria live on particles of food and nutrients in saliva. In the mouth, bacteria attach themselves to the teeth to form coatings or biofilms known as dental plaque. These biofilms are responsible for the two most common infections of humans—tooth decay, and gum and bone diseases. Dental diseases cost Australians many millions of dollars annually in dental services. More importantly they affect the health of millions of people, particularly children. Jacques' research career has been devoted to studying these oral bacteria.

- 4 How do humans get bacteria in the mouth?
- 5 (a) What is the biofilm that bacteria produce?
(b) What diseases do these biofilms cause?
- 6 Why is Nick's research into these bacteria important?

What did Jacques do?

While Jacques was doing his research, it became known that some bacteria in dental plaque are responsible for disease, but others exist harmlessly in our mouth. Some of the harmless bacteria even help prevent disease-causing organisms from living or taking over in the mouth. The way 'bad' bacteria attach themselves to teeth, the colonisation and development of biofilms on the teeth and gums, and the material they produce to cause disease has become a major area of international research.

- 7 What is the difference between the types of bacteria in the mouth (oral bacteria)?

GROUP WORK

- 8 (a) How would you study the different types of oral bacteria?
(b) How could you use these differences between bacteria in your research?

Recently it has been possible to isolate (separate) and clone (copy) the genes that give the instructions for the enzymes (biological chemicals) that form dental plaque. Jacques has studied a number of these enzymes in great detail in his laboratory to try and find out how they work, and how they can be stopped.

- 9** For successful research results Nick Jacques isolates and clones the genes of the bacteria.
- (a) What do these genes do?
 - (b) Why is it important to isolate and clone the genes?

What did Jacques conclude?

Until recently, Jacques' research concentrated on the enzymes that change sugar in our diet into large polysaccharide molecules. These polysaccharides are made from small molecules of simple sugars called monosaccharides. They form a matrix or cage that gives dental plaque its unique structure, and allow the disease-causing bacteria to reproduce and increase in numbers.

It has been established how these enzymes make the sugars fructan and glucan, and research will now concentrate on how to prevent the enzymes from working properly. This will then allow control over the amount of polysaccharide produced in dental plaque. In turn, this will restrict the development of dental disease.

- 10**
- (a) What makes dental plaque unique?
 - (b) What is a polysaccharide?
 - (c) Where do these polysaccharides come from?

Recently Jacques' team has learned how to monitor (or watch) oral bacteria, especially the genes which begin working once the bacteria attach to a tooth to form dental plaque. It is important to understand these gene changes, because bacteria that are in the biofilms have different properties and do different things to those that are found 'free-living' in the mouth. The biofilm bacteria are more likely to cause disease. The bacterial biofilms are also more difficult to kill with antibiotics, although at the moment it is not known why.

- 11**
- (a) What does monitor mean?
 - (b) Why is it important to monitor the oral bacteria?
 - (c) What are two differences between free-living bacteria and those in biofilms?

What happens next?

The study of how genes work in bacterial biofilms is an exciting new area of research that has been made possible due to recent developments in technology. This technology has come from many areas of science including research in chemistry, physics, molecular biology and information technology (IT). Study of biofilms will lead to a special understanding of how bacteria cause disease, not only in the mouth, but in more serious life-threatening bacterial infections in the body.

GROUP WORK

- 12** List some recent developments from research in chemistry, physics, biology and IT.
- 13** What sort of question could you ask Nick about his research?



Bacteria is the common name for the group of one-celled (unicellular) microscopic organisms that have a very simple structure. (Bacterium is used for a single unicellular organism.) Bacteria are thought to have been the first type of life on Earth. In fact, the oldest sign of life found so far is a fossil bacterial cell which was discovered in a rock in Africa. This cell is thought to be about 3 billion years old. Many of today's bacteria are useful to humans, but some are very dangerous pathogens and cause many diseases. Medical microbiologists study these pathogens, or micro-organisms that cause infectious diseases, in humans and other animals. In this way they hope ultimately to reduce the effect of the disease.

- 1 (a) How many cells in a bacterium?
- (b) When did the first bacteria appear on Earth?
- (c) What does a microbiologist do?

LOOK IT UP 🔍

- 2 (a) List some uses or benefits of bacteria.
- (b) List some diseases caused by bacteria.

The length of a bacterium is measured in nanometres.

1 metre (m) = 1 billion nanometres (nm).

Most bacteria range in size from 0.02 nm to 5 nm in length. Bacteria are found everywhere on Earth, including the outer edge of the atmosphere, the depths of the ocean, in frozen soil, and attached to rocks in hot springs or close to volcanoes.

The first bacteria were observed in 1676 by Anton van Leeuwenhoek. He made simple microscopes to observe bacteria as well as other microbes such as protozoa, yeast, and one-celled algae. Leeuwenhoek (1632–1723) is regarded as the father of bacteriology.

- 3 (a) How many nanometres in 1 metre?
- (b) 1 centimetre?
- (c) 1 millimetre?

LOOK IT UP 🔍

- 4 Bacteria live where other organisms cannot. Suggest some reasons why they can do this.
- 5 (a) Describe any differences between protozoa, yeast, and one-celled algae.
- (b) Describe any similarities between protozoa, yeast, and one-celled algae.

Bacteria are classified into three main types according to their shape:

- i **bacilli** — rod-shaped bacteria; they often have small whip-like structures called flagella used for movement. The diagram at the top of this page shows a bacillus bacteria.
- ii **cocci** —spherical bacteria
- iii **spirilla** —spiral-shaped; have a crescent moon shape.

The cocci and bacilli, in particular, can form long chains.

- 6 (a) What is a flagella?
- (b) Draw diagrams of bacilli, cocci and spirilla.

Some bacteria need oxygen to survive. These are called aerobic bacteria. Other bacteria, called anaerobic bacteria, can only live without oxygen. These live in oxygen-poor environments, for example near underwater volcanoes. Some of these bacteria use sulfur as a basis for food production.

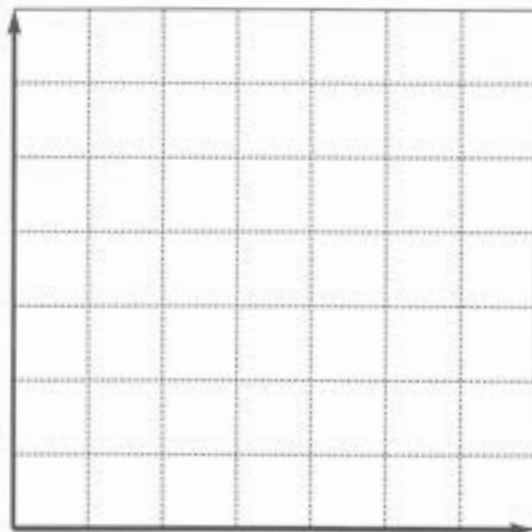
- 7 (a) What does the term 'aerobic' mean?
(b) What does the term 'anaerobic' mean?

Most bacterium reproduce by splitting into two new cells. Some bacterium can divide every 20 minutes. This fast reproduction explains the rapid development of disease and decay.

- 8 (a) Complete the table below to show how many bacteria are present if a single bacterium divides every 20 minutes.

Time	20	40	60	80	100	120	140
No.	2	4	8				

- (b) In the grid show the information from the table on a graph. Label time on the X (bottom) axis and the number of bacteria on the Y (upright) axis.
- (c) If I start with a bacterium which can divide and reproduce every 20 minutes, use the graph to find out how many bacteria I will have after:
- 1 hour?
 - 2 hours?
 - 3 hours?



Unless food is prepared and stored properly it can be spoiled by bacteria. Sometimes the food looks, smells or tastes 'bad' which warns us not to eat it. Sometimes we cannot tell it has been poisoned by bacteria and the poisons (toxins) can cause food poisoning. Two examples of dangerous diseases caused by bacteria in food are called salmonella and botulism. Food poisoning can be caused by food being left uncovered (and exposed to airborne bacteria); being kept too warm because it was not refrigerated; or by bacteria from our hands causing infection.

GROUP WORK

- 9 How can your school lunches be prepared and kept so that you reduce the risk of bacterial infections? List as many steps as you can in the process.

Bacteria are transmitted by air, insects, water, food, and direct contact with human beings, animals, and objects. Boiling water can destroy most bacteria in water. Disinfectants can kill bacteria on surfaces such as toilet seats or kitchen benches. Most disease-producing bacteria in milk can be killed by pasteurisation (named after Louis Pasteur). Antiseptics can kill bacteria, and freezing can reduce their growth. (Once the item of food melts, however, the reproduction of bacteria starts again!) Chlorine is commonly used to purify drinking water and in higher concentrations in swimming pools.

GROUP WORK

- 10 (a) What is your favourite meal?
(b) Describe how a professional cook should prepare your favourite meal so that it is protected against infection by bacteria.

Robert Koch (1843–1910)

Robert Koch was born in a town in Germany called Klausthal-Zellerfeld on 11 December 1843. When he turned 18, Koch went to the University of Göttingen and studied mathematics, physics and botany. He spent a small period of time at the Hamburg General Hospital and then started his own private practice in medicine. During this time Koch also studied anthropology, archaeology, occupational diseases and the new field of bacteriology.

1 Describe what activities you would be involved in if you worked in the following areas:

- (a) anthropology
- (b) archaeology
- (c) bacteriology.

THINK ABOUT IT

- 2**
- (a) What are some diseases that Robert Koch may have studied in the nineteenth century?
 - (b) What are some diseases of today?
 - (c) What can you do to reduce the effect of diseases in your workplace or at school?

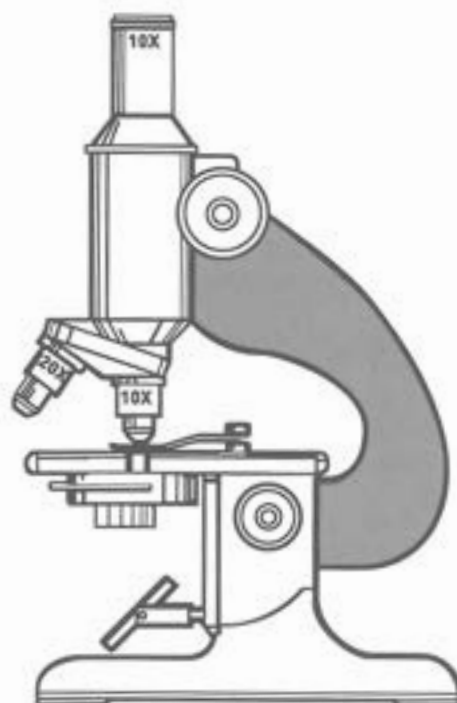
LOOK IT UP

- (d) Investigate what is being done to limit the number of cases of one of these twentieth century diseases.

The story so far

Glass lenses had been used since the early thirteenth century to magnify objects. It was not until the seventeenth century that microscopes were developed which are similar to those we use today. In 1676 Anton van Leeuwenhoek (1632–1723) discovered the first unicellular (single-cell) organisms, and he later discovered that bacteria were found in saliva. Many discoveries were made in the nineteenth century, and with Koch's medical background he developed an interest in the newly developing field of bacteriology.

- 3**
- (a) What scientific instrument do we use to study cells?
 - (b) Label the diagram as accurately as you can.
 - (c) Where do you put the cells you are observing?
 - (d) What magnification does the microscope have on the
 - i eyepiece lens?
 - ii objective lenses?
 - (e) How do you focus the microscope?



What did Koch do?

Koch discovered that it was bacteria that caused some diseases, and individual bacteria caused individual diseases. This discovery was made when Koch showed that if bacteria were isolated from anthrax-infected mice, and then injected into a healthy mouse, then that mouse would also contract anthrax. In addition, the same strain of bacteria could then be isolated from the newly injected mouse. After further studies of infections in wounds, Koch also isolated the bacteria responsible for tuberculosis.

In 1883, cholera reached epidemic levels in India. Koch travelled there and discovered that the disease was contracted mainly through contaminated water. He later travelled to Africa where he conducted research which showed that animals were often vectors or carriers of a number of major diseases.

THINK ABOUT IT

- 4 (a) Why is it important for scientists to know what causes a disease?
(b) Have you had an infection?
(c) If so, describe the symptoms (what the disease was like).
(d) How did you cure your infection?

What happened next?

Further research has been conducted on the anthrax disease. Scientists have devised ways in which the bacteria called *Bacillus anthracis* can be isolated and identified. They have also made a vaccine that can be administered via three injections (each injection is given at intervals of two weeks), followed by annual booster shots.

A vaccine is usually a modified virus which can stimulate the body's response to a disease. Many of the other diseases that Koch isolated have also been studied further and many of them now have a vaccine.

- 6 (a) Describe how a vaccine works.
(b) List some vaccinations that you may have had.
(c) What are booster shots and how do they work?

Some people think that anthrax may be used in biological warfare. The US military forces have vaccinated their soldiers, and there are about 10 countries that are able to load weapons with spores of *Bacillus anthracis*.

GROUP WORK

- 7 List the disadvantages of countries using biological warfare.
- 8 Write a "Letter to the editor" stating your views on biological warfare.



Robert Koch contributed greatly to our understanding of disease. From his research he developed four statements or postulates which describe how to search for the cause of a disease and how to determine which micro-organism causes a disease.

The postulates are:

- 1 The micro-organism must be present in every case of the disease.
- 2 The micro-organism must be separated or isolated from the diseased individual and then grown in a culture away from its host.
- 3 When a healthy host is exposed to the micro-organism it will develop the disease.
- 4 The micro-organism must be separated or isolated from the **new** diseased individual and then grown in a culture away from its host. This new culture must be of the same organism as before.

- 1 Rewrite Koch's Postulates in your own words.

LOOK IT UP 🔍

- 2 Write definitions for the following terms:

- (a) micro-organism
- (b) bacteria
- (c) fungus
- (d) mould
- (e) contamination
- (f) safety precautions

GROUP WORK 🗨️

- 3 How can we test Koch's postulates? [Follow the steps below.]

- (a) In your group describe the steps you would take to separate the common bread mould.
- (b) List any safety precautions you should follow to protect yourself.



- (c) What precautions would you need to take to stop confusion with other micro-organisms in the air contaminating the experiment?
- 4 Koch cultured micro-organisms. How could you culture the micro-organisms that cause bread mould?

Stopping the Spread of Disease

Robert Koch was very careful in his experiments not to allow cross-contamination from other sources, and he was also very careful not to contaminate himself with the micro-organisms he was studying. He used careful cleaning and kept accurate records.

In disease control laboratories today, the possibility of contamination is so dangerous that great expense is needed to protect the workers and the outside environment. We also need to protect our environment, and our bodies, against contamination by the ordinary micro-organisms which inhabit our world. Some of these are found in the air, or in water, or in the food we eat.

GROUP WORK

- 5 [a] What sort of contamination do we have to protect ourselves against?
- [b] All of us have caught the common cold virus. How can we protect ourselves and reduce the chance of catching a cold?
- [c] i List some other diseases in our environment.
- ii How can you protect yourself against the micro-organisms involved in these diseases?

Protecting Our Own Bodies

When preparing food or eating, the first thing we should do is wash our hands. This removes micro-organisms from the skin. Washing is especially important after using the toilet, gardening, or playing with pets and other animals. Some foods such as milk, cream or meat dishes need to be refrigerated in covered containers. Refrigeration deprives micro-organisms of warmth and a cover prevents airborne micro-organisms from settling on food to grow later. Fruit and vegetables (eaten raw) need to be washed to remove micro-organisms (and chemicals) from the skin of the fruit or vegetable.

Meat needs to be carefully prepared as it often contains bacteria. Steaks only need to be cooked on the surface, but minced meat and sausages need to be thoroughly cooked all the way through. This is because any bacteria on the original surface of the meat has been mixed through during the mincing process. The amount of time involved in this preparation allows for more bacteria to grow. Chicken contains many different bacteria and so should be cooked all the way through. Correct preparation of foods can prevent food poisoning as a result of bacteria.

- 6 [a] Why should you wash your hands:
- i before preparing food?
- ii after using the toilet?
- [b] What are some health problems associated with playing with pets and other animals?
- 7 You are about to have a snack which consists of a cheese sandwich and an apple.
- [a] What are some precautions you should take to protect yourself from micro-organisms?
- [b] Describe how you would prepare a sandwich for later the same day.

GROUP WORK

- 8 [a] What precautions would you take when barbecuing meat?
- [b] What could be some dangers of eating uncooked mince meat or chicken?

Jocelyn Laurence was born in the NSW country town of Cooma in 1976. She attended PLC in Sydney where she gained her HSC, having studied Physics, Chemistry, Maths, English and Geography. At Sydney University Laurence studied physics, as well as history and philosophy of science. In 1998 she completed her Honours BSc, looking at the health effects of electromagnetic fields. Laurence followed this with PhD research on the same topic.

- 1 List some thoughts you have about a possible career. What subjects will you need to study for this career path?

The story so far

All of us are continually surrounded by electromagnetic radiation from a range of different sources. Some of these sources are natural, including the sun. Others result, however, from technological advances which have occurred over the last century and, in particular, the last decade. Some examples include television transmissions, the use of high-voltage power lines, and extensive radar use for airports, seafaring and mobile phone telecommunications. In each case, low-level radiation continually passes through all of the cells in our body.



Since World War II, scientists have been gradually made more aware of the dangers of high-level radiation. The possible effects of non-ionising radiation (low energy radiation such as radio waves and microwave communications) are less well known. As a result of this lack of understanding, there is some fear in the community regarding the use of these waves and the extent of radiation in our lives.

- 2 List some dangers associated with radiation emitted from nuclear explosions.
- 3
 - (a) Do you use a microwave or a mobile?
 - (b) Are you exposed to radio and TV waves?
 - (c) List some different types of low-level radiation to which you are exposed.

GROUP WORK

- (d) List some of the concerns that your group has about low-level electromagnetic radiation.

What did Laurence do?

During her experiments Laurence found that, as radiation was absorbed in cells, it was changed into heat. Although the temperature change was very small it still caused stress in the cell. A protein in the cell changed shape as a result of this stress. Laurence's idea was that exposure of cells to radiation caused changes in the structure and shape of the protein, and that the amount of change was determined by the amount of energy absorbed by the cell.

- 4
 - (a) What activities done by you might cause your cells to be exposed to heat stress?
 - (b) How could you detect these changes?

Laurence decided to study this change by using both high-powered light and electron microscopes. Electron microscopes use beams of electrons instead of light to 'see' extremely small objects such as single molecules. The electrons are focused by magnetic lenses onto film or a digital collector in the same way light is focused using glass lenses.

- 5 Describe occasions when you use lenses to magnify (make larger) or diminish (make smaller) different objects?

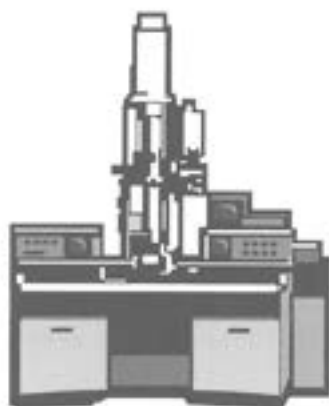
One aim of Laurence's study was to see if shape changes in the protein could be used to determine a safe radiation level. The first stage was to grow cultures of cells and subject them to radiation. The two frequencies Laurence used were 2450 MHz (similar to that found in commercially available microwave ovens) and 864 MHz (halfway between the frequencies used by digital and the (no longer available) analogue mobile phone signals).

- 6 (a) What frequencies of radiation did Jocelyn choose for her experiment?
(b) Why do you think she chose these frequencies?
- 7 What are the advantages of using cells grown in a culture, rather than real animals, for these experiments?

The tools used in these experiments included both light microscopes and electron microscopes. Using light microscopy and fluorescent dyes, Laurence observed and described not only the changes in the shape of the cell, but also determined the different proteins which were present in the cell.

- 8 (a) How would a fluorescent dye be used to view a cell under a microscope?
(b) What would be the point in using a dye to study different proteins?

Electron microscopes can magnify objects up to 500 000 times. This is about 1000 times better than most light microscopes. Some electron microscopes can resolve individual atoms which are only 1/10 000th of a millimetre apart. Only the electron microscope can show the molecular changes in the protein itself.



GROUP WORK

- 9 (a) What are some advantages of the electron microscope over the light microscope?

THINK ABOUT IT

- (b) Can you think of some disadvantages of electron microscopes?

What did Laurence conclude?

At this stage in Laurence's research, she (along with her co-workers P. W. French and D. R. McKenzie) developed a model for describing how electromagnetic radiation changed biological systems or living cells. This research was presented in November 1999 at a Symposium at the University of Sydney.

THINK ABOUT IT

- 10 (a) What sort of 'model' might Jocelyn have presented at the Symposium?
(b) What is a 'Symposium' and how could it help scientific research?
(c) What are some advantages of scientists working together rather than on their own?

What happens next?

The next step will be decided as the research progresses. Perhaps Laurence will show that current safety standards are acceptable, or her research may show there is a need to reassess standards. Laurence will certainly add to knowledge about how cells and proteins react to the stress of electromagnetic radiation. She also expects to extend the usefulness of electron microscopy as a tool to study molecule shape.

- 11 What sort of questions could you ask Jocelyn about her research?
- 12 You can contact Jocelyn yourself to find out about her experiments. Her address is: The Department of Applied Physics, School of Physics A28, University of Sydney, NSW 2006.

Electron microscopes use rays or beams of electrons rather than light to allow us to 'see' very small objects. Like light being focused through lenses of glass or plastic, electrons are focused by using lenses of magnetic fields. The electron beams are focused by magnetic fields bending the path of the electrons through an angle, just as a glass lens bends the path of a light ray.

The electrons are transmitted through a vacuum to the object. They either pass through a very thin object in the case of a transmission electron microscope (TEM), or bounce off the surface of a larger specimen in a scanning electron microscope (SEM). In each case, the electrons are focused by the magnetic fields onto a photographic plate or a digital collector.

- 1 (a) How are electron microscopes and light microscopes similar?
- (b) How are they different?
- 2 (a) What is a TEM?
- (b) What happens to electrons in an SEM?
- (c) Describe one difference between a TEM and a SEM.

The electron beam passes along what is called an optical axis which is in the centre of ring-shaped magnets. In the diagram, the upper pole is north and the lower pole is south. The magnetic field B (shown by the lines with arrows) is curved due to the shape of the magnet's surface.

This curve of the magnetic field is an important part of the focusing process and maintains the path of the electrons. The image plane is where the electrons are received by a viewing camera or photographic film.

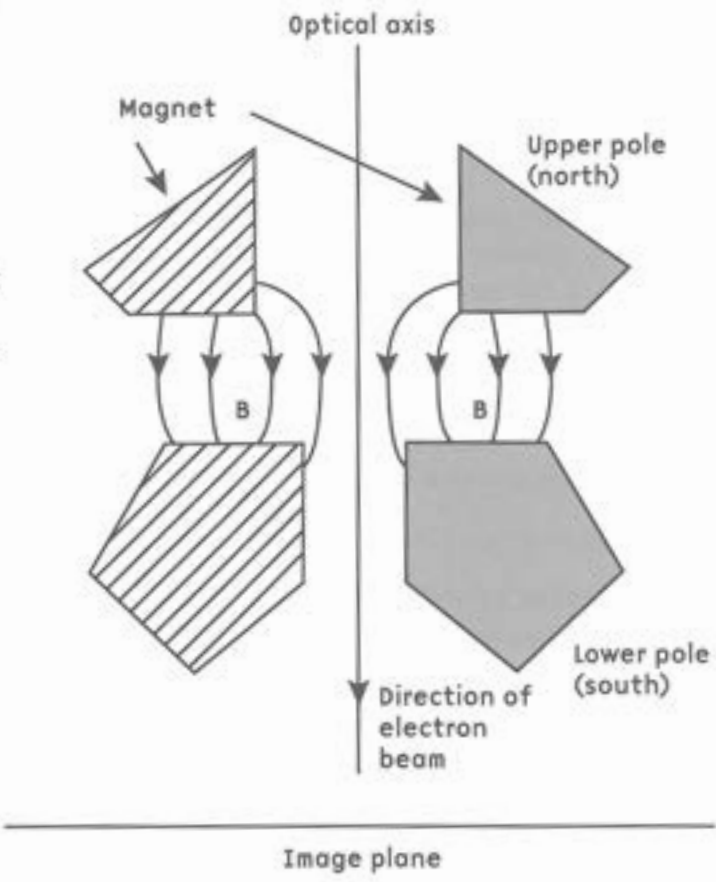
- 3 What is the purpose of the curve of the magnetic field?
- 4 Look at the diagram.
 - (a) What happens at the image plane?

THINK ABOUT IT

- (b) Why are the ring magnets at a different radius from the optical axis?

LOOK IT UP

- 5 What other machines used by you function or work because of the magnets they contain?

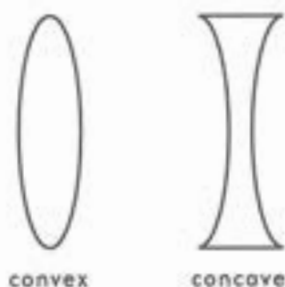


History of Microscopic Resolving Power

From earliest times, humans have wanted to see smaller and smaller objects. The ancient Greeks knew about the magnifying properties of drops of water and differently shaped pieces of glass. In the thirteenth century Robert Grosseteste first proposed the word lens (because these shaped pieces of glass were shaped like lentils). He also proposed that they could be used to observe objects which were too small to be seen easily.

The first lenses to be made regularly were convex lenses (the centre of the lens was thicker than the outer edge).

6 Using this diagram show how convex and concave glass lenses can focus light rays.



Anton van Leeuwenhoek (in 1675) ground and polished a small glass ball into a lens with a magnification of 270X, and became the first person to make and use a real microscope. This microscope had only one lens, and is called a single-lens microscope. Leeuwenhoek continued to observe bacteria, germs and sperm with some of the 400 microscopes he made during his life.

To increase the magnification (and resolving power) of a single-lens microscope is very difficult, and the compound microscope (using more than one lens) system was invented in the seventeenth century. This meant that the image magnified by one lens is magnified again by the next lens. One way that the magnification of a microscope can be described is by its resolving power. Resolving power is how close two objects can be and still be determined as two (rather than one) object. Electron microscopes have a much higher resolving power (about 10 000 times greater) than the resolving power of a light microscope. Magnification of a microscope can be described by resolving power which is how close two objects can be and still be determined as two (rather than one) object.

7 [a] What does resolving power mean?

[b] Explain what it means to have a resolving power of 10 000 times greater than that of a light microscope when observing a specimen?

Since its invention, the compound microscope has been used by Robert Hooke (1635–1703) to determine that living things are made of cells. Louis Pasteur (1822–95) of France used a compound microscope to discover yeast fungus and Robert Koch used one to observe the bacteria responsible for tuberculosis and cholera.

One of the most famous early pioneers of lenses in glasses was Benjamin Franklin, who is credited with inventing bifocal glasses in the late 1700s. In 1887, a German named Adolf Eugen Fick began to develop contact lenses.

GROUP WORK

8 [a] Do any of your teachers or relatives wear bifocal or multifocal glasses?

[b] What do you think are some advantages of wearing these type of glasses?

[c] List some advantages and disadvantages of contact lenses.

Antoine Laurent Lavoisier (1743–1794)

4a

Educated at the College Mazarin in Paris, Antoine Lavoisier studied mathematics, astronomy, chemistry, botany, and legal studies. His first job was to collect geological information for a map of France. Lavoisier's first published research was on the properties of the mineral gypsum. Later he was awarded a gold medal by the King of France for research on street lighting. Lavoisier worked in a number of areas including the Royal Gunpowder Administration and established the metric system of weights and measures. In 1793 he was arrested by the leaders of the French Revolution because of his involvement in tax collection. Lavoisier was executed by the guillotine.

- 1 (a) What benefits would street lighting have that would lead to Lavoisier being awarded a medal?
- (b) Before Lavoisier's time, France had many types of measurements. What advantage does the metric (SI) system have over other types of measurements?
- (c) Suggest a reason why Lavoisier was executed by the revolutionaries.

The story so far

In 1597 the first chemical textbook *Alchemia*, by Andreas Libavius, was published. Alchemia or chemistry was defined as the art of 'producing reagents and extracting pure essences from mixtures'. Many new compounds were being prepared at this time by experimenting with animal and vegetable materials. Some of these were alcohol, arsenic, zinc and three strong acids—hydrochloric, sulfuric and nitric. At this time Georg Stahl (1660–1734) proposed the Phlogiston Theory as a way of explaining what happens when substances burn or react with air. Substances that burnt easily such as coal were thought to contain large amounts of phlogiston. (Phlogiston does not actually exist.)

GROUP WORK

- 2 (a) How could you obtain compounds from living material such as a leaf?

LOOK IT UP

- (b) Write the modern formulas for:
 - i hydrochloric acid
 - ii sulfuric acid
 - iii nitric acid
 - iv arsenic
 - v zinc
 - vi alcohol



What did Lavoisier do?

Even though Lavoisier is regarded by many as the founder of modern chemistry, he did not actually discover any new elements or compounds. He did however describe many experiments. In 1772, Lavoisier began a series of experiments that demonstrated the nature of combustion (burning), and proved that the Phlogiston Theory was wrong. From these experiments he concluded that combustion results from the union of a flammable material with the newly discovered gas, oxygen. (Although oxygen had been discovered by Joseph Priestley a few years before, Lavoisier named it and showed that it made up about 20% of the atmospheric air.) Lavoisier accurately presented the chemical knowledge of his time in a chemistry textbook called *Elements of Chemistry* which was published in 1789.

- 3 Describe what you think will happen when the following materials are burned:
 - (a) match
 - (b) match head
 - (c) methylated spirits
 - (d) kerosene
 - (e) wax

4 Some of these materials produce smoke. What is in smoke?

- 5 [a] What main gases make up the atmosphere?
[b] What are some uses of oxygen?

Lavoisier was very careful in his measurement of the weights of substances involved in chemical reactions. He heated mercury oxide and showed that it released oxygen. The material weighed less after the oxygen had gone. Lavoisier also showed that if mercury was heated at a lower temperature, it would react with oxygen and weigh more.

- 6 [a] Matter never disappears or appears in chemical reactions. Why does mercury [Hg] weigh more when it is heated with oxygen [O] to form mercury oxide [HgO]?
[b] How much carbon dioxide [CO₂] is formed in the following reaction between carbon [C] and oxygen [O]?



What did Lavoisier conclude?

Using the ideas he developed about the composition of the atmosphere, Lavoisier described what happened in the combustion process in chemical reactions, and what occurred during respiration in living cells. Because of his work we now know that respiration involves the slow, controlled burning of sugar with oxygen in the cells of living things. Respiration also results in the release of the waste product carbon dioxide.

- 8 [a] What is respiration?
[b] What raw materials are used in respiration?
[c] What is the main waste product?

As a result of these experiments, Lavoisier defined a chemical element as a substance that could not be decomposed into more simple substances by heat or any chemical reaction. He also defined a compound as a combination of two or more elements in a fixed and definite proportion by weight. Each element was given a number, called combining weight, and it was this number (or any whole number multiple of it) that represented the amount of the element which took part or combined in a chemical reaction.

He developed the first list of 33 elements. This list included lime and magnesia which we now know are not elements. These two substances were later studied by Humphry Davy (1778–1829), who separated the elements calcium and magnesium from them.

- 9 [a] What is an element?
[b] Give two examples of elements.
[c] What is a compound?
[d] List two compounds.
[e] What elements make up these compounds?

What happened next?

This idea of basing chemical reactions on weight placed chemistry on a quantitative, or measurement, basis. Prior to this chemistry had been qualitative or descriptive only. Lavoisier's work was the start of atomic theory and led, in time, to the chemistry we now know.

- 10 [a] What does the word qualitative mean?
[b] What does the word quantitative mean?
[c] How do you think chemistry is a quantitative science?

Mixtures

Most of the matter that we see around us is mixtures of different substances. Two examples are tap-water and soil. Other examples include blood, tree sap, urine and sea water. Most of these are naturally occurring and we produce some of them in our own bodies. Chemists who worked with Lavoisier would have studied the local drinking water and soil to see what they could find out about these mixtures of substances. Many of the chemicals that they used came directly from these two mixtures. Other mixtures they studied were milk, olive oil and wine.

- 1 (a) What products were studied by the chemists who worked with Lavoisier?
- (b) Why did the scientists of the eighteenth century mainly study the natural products mentioned?
- (c) What differences are there between different types of milk or between different wines?

Tap-water contains mainly water and trace amounts of different salts such as calcium and sodium compounds, as well as salts containing chlorides and fluorides. It also contains some very small particles of mud. Tap-water taken from Sydney will be different from that taken from Adelaide or Perth. Being a mixture means that all the different substances found in ordinary tap-water have not joined to make one substance but are just found together.

- 2 (a) List some other examples of common mixtures.
- (b) List some common compounds.
- (c) What other things might be present in tap-water?
- (d) When we travel to some areas of the world, it is not safe to drink the water. Why?

Soil is a mixture made up mainly of solids. Some of the solids are large (small rocks and stones), but others such as bacteria and fine grains of clay, and water molecules in the spaces between the particles of soil, are very small. Like tap-water, soil contains different salts and other compounds and may even contain elements such as nitrogen (as part of the air found between the soil particles). Many soils contain living things and chemicals that used to belong to living things, such as decaying bodies of insects or faeces.

GROUP WORK

- 3 Look at some examples of soil using magnification.
- (a) What is present in soil?
- (b) Draw what you see through a hand lens and a binocular microscope. If you see any living things in the soil, draw these as well.

Compounds

When substances are found together all the time in specific numbers or proportions, they are called compounds. Elements react together to make compounds and they can only be separated by decomposition. A compound is called a pure substance because we know exactly what is in it. Even though our tap-water is safe to drink, by chemical definitions it is not a pure substance.

Many compounds are called salts. The common salt that is often found in snack foods is sodium chloride and it is a compound of sodium (Na) and chloride (Cl). Sodium bicarbonate is a type of salt, and so is blue copper sulfate which is often used in science classes. In each case the amount of each element (shown by the different letter symbols) is always exactly the same. Other common compounds are sulfuric acid and calcium hydroxide.

LOOK IT UP

4 Write the name for the following compounds:

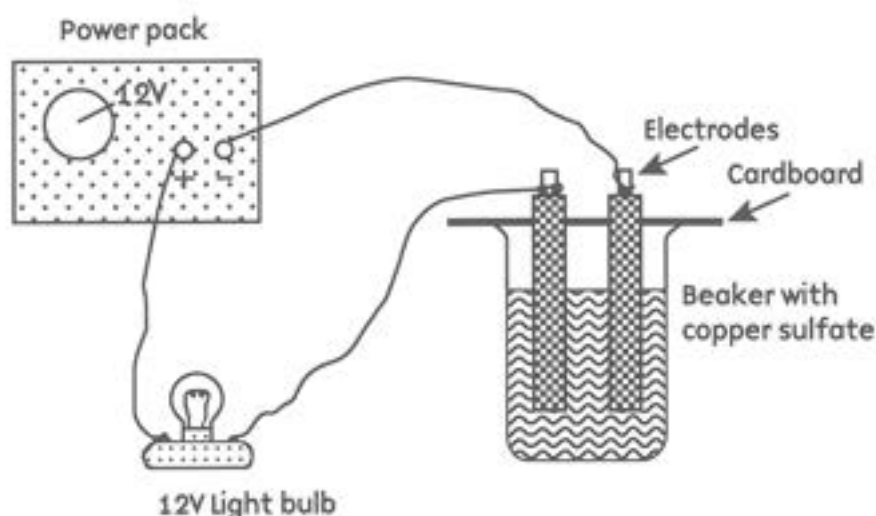
- (a) NaCl
- (b) NaHCO₃
- (c) CuSO₄
- (d) H₂SO₄
- (e) Ca(OH)₂

Some compounds are very difficult to decompose but others can be decomposed in the laboratory. Copper sulfate (CuSO₄), for example, dissolves in water to form a bright blue solution and the copper (Cu) can be collected by using electricity. The following activity shows how.

GROUP WORK

Materials

medium beaker
copper sulfate solution
two electrodes
connecting wires
12V light
transformer rectifier



Method

Half fill a beaker with copper sulfate solution. Place electrodes in the solution but not touching each other. (Use a piece of cardboard with two holes and insert one electrode in each hole. This should stop them touching each other.) Connect the bulb, power pack and electrodes as shown in the diagram.

5 When you turn on the power, describe what you see in each of the following places:

- (a) light bulb
- (b) electrode connected to the positive (+) terminal
- (c) electrode connected to the negative (-) terminal
- (d) in the solution or at the bottom of the beaker.

6 Describe what is happening at:

- (a) light bulb
- (b) electrode connected to the positive (+) terminal
- (c) electrode connected to the negative (-) terminal
- (d) in the solution or at the bottom of the beaker.

7 (a) What have you done to the copper sulfate compound?

- (b) How can you tell that copper sulfate is a compound?

In the nineteenth century Charles Lyell's geology textbooks were considered necessary reading for all geology students. Lyell was born in Scotland but after studying law at Oxford, he spent most of his life wandering around the world, studying the geology of areas and trying to tell people about science. Lyell was successful, particularly through his *Principles of Geology* which was printed in 12 editions. The ideas he presented in this and other books shaped thinking, both in geology and other sciences, up to the present day.



- 1
 - (a) Are books useful in spreading information and knowledge?
 - (b) If you were Lyell and wanted to teach people about your favourite subject, how would you do it?
 - (c) Do you have a text book which is easy to read and interesting? If so, describe what makes it interesting.
 - (d) If you were able to travel the world and study something, where would you go? Why?

The story so far

One of the most popular geological theories of the day was the theory of catastrophism. This theory stated that most of the Earth's history consisted of a short period of violent geological activity. This idea was based on some people's interpretation of Creation described in the Bible. Bishop James Ussher (1581–1656) calculated that the Earth was created in 4004 BC. Because of this idea, most people believed the Earth had only been in existence a few thousand years. These people thought that violent earthquakes, volcanoes and floods had happened 'all the time', and created the world around them.

A totally different idea was developed by James Hutton (1726–97), a Scottish geologist. He thought the world had been made some millions of years earlier and that changes had occurred much more slowly. Lyell called Hutton's theory uniformitarianism and supported it in his writings as strongly as he could.

- 2
 - (a) What is the theory of catastrophism?
 - (b) What is uniformitarianism?
 - (c) What did Bishop Ussher do?
 - (d) How old did Ussher think the world was?

What did Lyell do?

In the first volume of *Principles of Geology* (published in 1830), Lyell argued that geological activities could be explained as 'what was happening now'. Lyell said that although volcanoes and earthquakes had been occurring for a long time, they were relatively few and located in different areas around the world. The second volume (1832) described the geological activities which occur regularly in the world. These include the action of rivers (in eroding areas of mountains and depositing layers of sediment), and volcanic activity (building mountains and producing new rock).

- 3
 - (a) What are the main ideas of Lyell's first two volumes of *Principles of Geology*?
 - (b) What main geological activities occur in the world?
 - (c) What is the difference between erosion and deposition?

In the second volume of *Principles of Geology* Lyell made up a number of new words to describe what was happening. One of these words was 'metamorphic'. This describes sedimentary rocks which have been changed by being close to volcanic activity, and been affected by heat or pressure or both. Even igneous rocks from previous volcanic activity can be metamorphosed or changed by new volcanic activity.

- 4** [a] What is a metamorphic rock?
 [b] How is one formed?
 [c] Could you find fossils in metamorphic rocks? Explain.

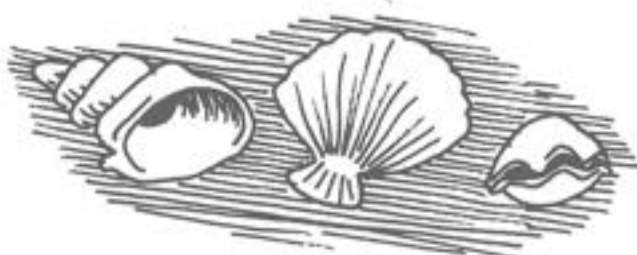
In Volume 3 (1833) Lyell described the ideas of 'stratigraphy' and 'palaeontology'. Stratigraphy describes how rocks are found in layers, and how the early rocks to be formed are on the bottom and later rocks are on the top. This was not Lyell's original idea. It had been developed by William Smith (1769–1839), an English geologist who studied fossils.

In 1816 Smith listed the strata or levels of rocks according to the types of fossils which he found. Palaeontology is the study of fossils. Both of these ideas help describe what has happened in the Earth's structure.

- 5** [a] What is the difference between 'stratigraphy' and 'palaeontology'?
 [b] What was William Smith's major idea?

What did Lyell conclude?

From his studies in geology and zoology (the study of animals), Lyell classified or grouped the rocks of northern Italy. Unlike fellow geologists, Lyell looked at the fossils rather than the rock types to help him date and describe the different layers. The fossils that he studied were shellfish and based on these he was able to date the rocks and describe how they had been made.



- 6** How could Lyell use shells of shellfish to date rocks?

Lyell also produced new terms to describe different geological periods (called epochs). These were Pleistocene, Pliocene, Miocene, and Eocene. These were Lyell's words which are still basically used today.

What happened next?

Charles Darwin read Volume I of Lyell's *Principles of Geology* before his famous trip on the Beagle in 1831. It is likely that Lyell's ideas on how long the Earth had been in existence, and how it had changed slowly but regularly, helped Darwin develop his theory of evolution. Lyell also helped Darwin and Alfred Russel Wallace (1823–1913), another philosopher who considered the origin of the Earth and life on it, to publish their ideas.

GROUP WORK

- 7** [a] For what is Charles Darwin famous?
 [b] How could his ideas be helped by the writings of Lyell?

Today we believe that the Earth is about 7000 million years old and although we still don't know exactly how it was formed there are many ideas. One idea has the Earth forming from a huge cloud of dust and rocks revolving around the sun. Gravity between the particles attracted them to each other making the cloud more and more solid. This forcing of particles together increased the heat, and melted and fused all these particles into one huge mass of Earth. As the cooling and heating continued over time the main areas of the Earth — core, mantle and crust — formed from different types of material and different rates of cooling.

1 What is one idea about the formation of the Earth?

The Earth's history after it cooled (geological history) is often divided into four main sections or eras.

Proterozoic Era (4600 to 600 million years ago)

This time section began about 4.6 billion years ago when the first crust was forming. The Earth's surface was very hot and water escaping from the crust (in the form of steam) formed large rain clouds. Gradually, the crust cooled and thickened. About 250 million years later, the crust was cool enough not to evaporate the rain from the clouds. From this, the first oceans and seas were formed.

In this era, the first organisms (very similar to bacteria or single-celled algae) appeared in the oceans. Due to weathering and erosion from air and water, the first soils appeared on land. As the algae and bacteria released oxygen from photosynthesis, some of it reacted with elements in the crust (such as iron) and some was added to the atmosphere.

The main period of life in this era is called the Pre-Cambrian and it contains the earliest known fossils which are simple worm-like animals and single-celled organisms such as bacteria. Very few fossils were formed as the organisms had soft bodies. The oldest known Pre-Cambrian rocks are found in South Africa but the oldest evidence of life is found in the Pilbara region of Western Australia.

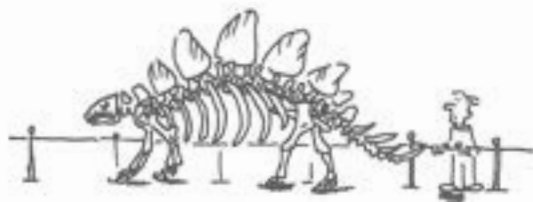
- 2**
- (a) What happened to the Earth to allow rain to occur?
 - (b) Draw a labelled diagram of a typical organism from the Pre-Cambrian era
 - (c) Why is it difficult to find fossils from this era?
 - (d) Where was the oldest life in the world found?

Palaeozoic Era (600 to 220 million years ago)

The Palaeozoic Era contained different periods of life and we can find a large number of marine fossils from this time, many of which had hard shells of calcium carbonate. These shells were made possible by changes in the concentration of carbon dioxide in seawater. These calcium carbonate shells are similar to shells found today. Other species of animals and plants appeared during this era. The first animals with backbones (vertebrates) were fishes and these are found, along with amphibians and reptiles, in later periods. Land plants evolved during this time to form large swampy forests. It is from carboniferous forests that we mine a lot of the coal we use today.

- 3**
- (a) Draw a diagram to show the backbone of a vertebrate animal, such as a fish.
 - (b) Explain how shellfish started to make calcium carbonate shells.
 - (c) What does 'marine' mean?

Mesozoic Era (220 to 70 million years ago)



The animals and plants in this era are very different from those of previous eras. The most obvious difference is the presence of dinosaurs. Other changes include the first birds, mammals and pine trees. Flowering plants and trees similar to the ones we see today appeared towards the end of the era. In the sea were many forms of corals and other kinds of shellfish.

When the surface had finally become solid in the Proterozoic Era, there was one huge continent called Pangea. In the Mesozoic Era it began to break up and separate into the continents we recognise today. Because of this land breakup, many types of land animals developed separately in the continents. This movement was first suggested by the geologist Alfred Wegener (1880–1930) and is called 'continental drift'.

- 4 (a) How are dinosaurs different from animals we have on Earth today?
- (b) What are some similarities between dinosaurs and animals today?
- (c) List some differences between flowering plants and pine trees.
- (d) What did Alfred Wegener propose in his theory?

Cainozoic Era (70 million years ago to present day)

At the beginning of this era great changes to the Earth's crust occurred and many mountain chains such as the Rockies in North America, Andes in South America, Alps in Europe and Himalayas of Central Asia were formed. All this movement was due to the drifting and collisions of the continents, along with the formation of deep rift valleys in Africa and other areas of the world. Perhaps it was these changes, or the collision with a large meteor, which wiped out the dinosaurs and many other organisms at the start of the Cainozoic Era. Certainly the fossils from this era are more similar to present-day animals and plants than earlier organisms.

The last three million years of the Cainozoic is known as the Quaternary period. Fossils which describe the evolution of humans are found in rocks of this age. We had arrived and we have been changing the world ever since.

- 5 (a) What major geological change occurred at the beginning of the Cainozoic Era?
- (b) How do you think the dinosaurs died out?
- (c) Describe any scientific evidence you have for your suggestion.

GROUP WORK

- 6 With your group make a time map for the Earth. You will need a roll of toilet paper (500 sheets), marking pens and tape.
 - (a) Assume the age of the Earth is 5000 million years. How many million years per sheet of toilet paper?
 - (b) Draw a line at one end and mark 'Present Day'. Mark the other end 'First Life'. What organisms will you draw here?
 - (c) Mark the Eras on the roll. Include animals and plants that were present at various stages.
 - (d) Which was the longest and the shortest Era?
 - (e) Explain the differences between the lengths of the different Eras.

Barbara McClintock was born in 1902 in Brooklyn, New York. As a student, she studied botany at Cornell University graduating with her PhD in 1927. She later taught at Cornell before becoming a Professor at the University of Missouri in 1936. Between 1942 and 1967, she did research on cytology (study of cells) and genetics at a research laboratory in Cold Spring Harbour, New York. In 1983, McClintock was awarded the Nobel Prize in Physiology and Medicine, one of only six women to have won this.

- 1 [a] What is genetics?
[b] What is botany?
[c] Explain how someone could study cytology, genetics and botany?
- 2 [a] What type of people would win the Nobel Prize for Medicine and Physiology?
[b] The Nobel Prizes have been awarded since 1901. Why do you think there have been so few women recipients?

The story so far

Gregor Mendel (1822–1884), the father of genetics, developed ideas on tracing characteristics from parents to offspring. He was a monk who from 1851 to 1854 studied the characteristics of pea plants in the monastery vegetable garden. In Australia, the first advances in practical genetics were made by William Farrer, with the development of wheat strains that could withstand the fungal infection called rust.

- 3 [a] Who was Gregor Mendel?
[b] Why is he called the father of genetics?
[c] What does the word 'characteristic' mean?
[d] What are some characteristics of different plants that you know?
[e] What are some characteristics of different animals that you know?
- 4 [a] Why was disease-free wheat so important for Australian farmers?
[b] How could the work of Gregor Mendel help William Farrer?



Other scientists did not notice Mendel's research results until 1900. In this year, three biologists, Hugo de Vries from the Netherlands, a German botanist Karl Correns and an Austrian Erich von Taschermak, separately obtained results similar to Mendel's while undertaking their own research. Also in the early 1900s, Thomas Hunt Morgan (who won a Nobel Prize in 1933) and his co-workers from Cornell University showed that genes or characteristics were on chromosomes, and they made the first genetic map.

- 5 [a] What was important about the work of de Vries, Correns and Taschermak?
[b] What is a genetic map?
[c] Where are genes found in the cell?

What did McClintock do?

Like Farrer, McClintock studied both cytology and genetics and how they related to agriculture. Her first major contribution to genetic research was made when she learned to identify each of the 10 chromosomes in corn. She was responsible for most of the contributions made to genetics by the Cornell genetics group.



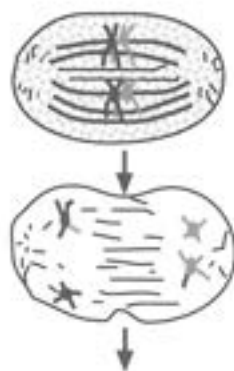
- 6** (a) Why is it important to study corn?
 (b) Are the characteristics of corn the same as those of wheat? Explain.

In 1931, McClintock followed Morgan's research and demonstrated that a process called crossing over (which is important in the production of egg and sperm cells) involves the physical exchange of chromosomal material (called genes). This exchange of genetic information means that offspring are not exactly the same as their parents. Today all scientists know that the different characteristics in cells are found on chromosomes and accept their physical structure.



Later in McClintock's career, when she was an Assistant Professor at the University of Missouri, she began to study chromosomes that had been broken by X-rays. McClintock found that the radiation changed (mutated) the chromosomes. Later she organised a way of using these chromosomes to produce new mutations. This led to McClintock discovering the process of 'transposable elements' for which she won the Nobel Prize.

- 7** (a) Crossing over is a special process in cells.
 i Where does it occur?
 ii What is the result of crossing over?
 (b) What is a mutation?
 (c) How can cells become mutated?
- 8** Often cells which mutate die. Why is this?



What did McClintock conclude?

The research on transposable elements could only be carried out by using special molecular experimentation that separated the parts of the chromosome which were moveable. These pieces of DNA which can jump in and out of chromosomes are sometimes called 'jumping genes'. In the late 1960s, jumping genes were obtained from a bacterium and called transposons. We now know that transposons (or jumping genes) may make up to 20% of a cell's genetic material.

- 9** (a) What is a transposable element?
 (b) What is DNA?
 (c) What is a transposon?

What happens next?

The next step is to find out if these transposons produce diversity or variations in natural populations of organisms. To do this scientists must observe genetic variation and measure the activity of certain enzymes in the cell. Already there is some research that shows transposons do cause genetic variation in natural populations. These jumping genes are presently being classified and their effect on cells is being studied at laboratories around the world.

- 10** (a) What is genetic variation?
 (b) Explain the terms 'diversity', 'enzyme' and 'population'.

Robert Hooke observed and named cells in 1665. His observations were followed by those of Anton van Leeuwenhoek (1632–1723), who described bacteria and protozoa. Today we know that all living material is made up of cells. All cells come from other cells usually by the process called cell division, but also by the joining of a sperm and egg cell. Every cell is contained inside a cell membrane and, although all cells are similar in their chemical makeup and size, there are differences between the cells that make up different organisms.

- 1 What parts did Hooke and Leeuwenhoek play in our understanding of cells?
- 2 List four important statements which describe our understanding of cells.

Parts of Cells

The cells that we are most familiar with — plant, animal and fungal cells — all contain a nucleus. They often contain other components, such as mitochondria and plastids. These small parts of the cell are called organelles. The genetic material is found in the form of chromosomes in the nucleus.

All cells must take in raw materials for their life processes (metabolism) and excrete waste products such as carbon dioxide. The raw materials are needed to make the molecules necessary for life or are used as fuel for energy. Metabolic functions and cell division require different organelles.



- 3 (a) What do plant and animal cells both contain?
- (b) Why do cells require raw materials?

Surrounding each cell is a thin cell membrane. This membrane controls movement of molecules in and out of the cell. Membranes are also found around each of the small organelles inside the cell. Around the outside of the cell membrane of plant and fungus cells there is also a cell wall, and this helps the cell keep its shape. Animal cells do not have a cell wall.

- 4 (a) What is the purpose of the cell membrane?
- (b) Why is a cell wall useful in plant cells?
- (c) Why are animal cells better off without a cell wall?

GROUP WORK

- (d) Some animal cells have set shapes without a cell wall. Suggest some reasons for this.

Organelles

- 5 Draw a diagram of each organelle in the space provided in the table below. [Use the previous cell diagram to help you.]

Organelle	Description	Diagram
Nucleus	Most cells have a single nucleus. Genetic information is carried in strands of DNA contained within the chromosomes in the nucleus.	
Nucleolus	Ribosomal ribonucleic acid (rRNA) is made here. This large molecule is like DNA and is used to get the information from DNA (genes) to the rest of the cell. The information is used to make proteins.	

- 6 (a) What is genetic information?
 (b) What molecule makes up the genes?
 (c) Special proteins called enzymes run the activities in the cell. How are enzymes controlled by the DNA?

Ribosomes	Ribosomes are tiny particles made of RNA. This is where protein is made in the cell.	
Endoplasmic Reticulum	This is the name given to large areas of membrane that often contain ribosomes.	
Mitochondria	All cells use oxygen to produce energy. As a result of this, carbon dioxide is produced. Respiration occurs in the mitochondria.	

- 7 (a) Where are ribosomes often found?
 (b) What do ribosomes do?
 (c) Do all cells have mitochondria? Explain.

Chloroplasts	Photosynthesis is how plants get energy from the sun. Photosynthesis occurs in chloroplasts using a green pigment called chlorophyll.	
Vacuole	Large areas of plant cells used for the storage of nutrients and food supplies.	

- 8 (a) Are chloroplasts found in animal cells? Explain.
 (b) What is chlorophyll?

Cell Reproduction

During cell reproduction, the chromosomes (made of DNA) double and then split into two cells exactly the same as the original. This means that when your cells reproduce, for example to replace skin or to grow more muscle, the new cells are the same as the old. We continue to look and act the same from one day to the next. In the production of sperm and eggs, the chromosomes also swap bits of DNA with other chromosomes. This is called crossing over. When the gametes (eggs and sperm) are complete, each of them is slightly different from the others.

- 9 (a) What happens during cell reproduction?
 (b) What extra process happens in the production of gametes?

Tobolsk in Siberia (Russia) is a very cold place, and is where Dmitri Mendeléev was born, the youngest of 14 children. Just as he finished school, Mendeléev's father died and the family's glass factory burned down. Mendeléev's mother took him to St Petersburg where he studied science, graduating in 1856. Mendeléev was appointed a Professor of Chemistry from 1866.

In 1869 there were 63 known elements (today there are more than 103). Mendeléev arranged these elements into a periodic table based on atomic mass. He also predicted the discovery of new elements based on gaps in his periodic table. It is for the development of this table of chemical organisation that Mendeléev is best known. He was also considered one of the great teachers at the University of St Petersburg. In 1890 Mendeléev resigned his professorship as a protest against political censorship and in 1893 he became Director of the Bureau of Weights and Measures.

- 1 (a) How many elements do we know today?

LOOK IT UP 🔍

- (b) Most of the recently discovered elements were made by nuclear reactions. List five of these.
(c) List some elements which would have been known by Mendeléev.

The story so far

In 1829 the German chemist Johann Döbereiner (1780–1849) developed the Theory of Triads, based on the chemical reactions of the elements known at that time (about 40). Some elements he put together were lithium, sodium, potassium (very reactive metals), and chlorine, bromine, iodine (very reactive and coloured gases). Each of these groups, as well as looking similar, had very similar chemical reactions.

In 1860, the English chemist John Newlands (1838–1898) developed a more extensive Law of Octaves. In his Law, the elements were listed according to increasing atomic mass or weight. Newlands proposed that the similar structure of the atom and chemical reaction is repeated after every eight elements. (For example Element 1 will have similar properties and react in a similar way to Element 9 and Element 17.)

You will need a period table for these questions.

- 2 (a) Find lithium, sodium and potassium on the periodic table. What group (column) are they in?
(b) List the other elements in this group.
(c) Describe any similarities (from the information on the table) between these elements.
- 3 (a) Find chlorine, bromine and iodine on the periodic table. What group (column) are they in?
(b) List the other elements in this group.
(c) Describe any similarities (from the information on the table) between these elements.

What did Mendeléev do?

Mendeléev also arranged the elements in order of atomic mass. The elements (by now 63) were placed one under the other according to whether they had similar properties. Unlike Newlands, however, he didn't fill up the gaps when an element didn't seem to fit but left space. For example, there were gaps underneath aluminium, boron and silicon. Mendeléev predicted the existence and properties of new elements to fit into these spaces. He also showed where accepted atomic weights for some known elements were incorrect. This organisation of elements was published a year before the work of Lothar Meyer (1830–1895), who was working on the same idea in Germany. In 1882 they jointly received the Davy Medal for their separate work.

- 4 (a) Find magnesium (number 12) in the periodic table. Name three elements that are in the same column.
(b) What is the name of the element 79?
(c) Find the elements which have atomic numbers of:
i 16
ii 92.

- (d) What element is immediately below aluminium?
 (e) What element would you expect to be similar to silicon?

What did Mendeléeve conclude?

The important feature of Mendeléeve's table was that he left space for new elements. One predicted new element was called eka-aluminium, because he believed it would have properties similar to that of aluminium. This element was actually discovered by Paul Emile in 1875, and he named it gallium, finding it to be similar to aluminium. Other elements predicted by Mendeléeve were eka-boron (discovered by Lars Fredrick Nilson in 1879 and named scandium) and eka-silicon (discovered by Clemens Winkler in 1886 and named germanium).

Today the modern periodic table lists all elements in atomic number order. The atomic number is the number of protons (positive particles) in an atom of the element. Protons and neutrons (neutrally charged particles) are the small particles which make up the nucleus of the atom. Around the nucleus are found negative electrons. Most periodic tables provide information about elements similar to that given for hydrogen in the box below.

- 5 List information for element number 3 (lithium), similar to that given for hydrogen.

Symbol	Atomic number	H	1	Li	()
Name		Hydrogen			
Date of discovery		1790		1818	
Number of protons, electrons, and neutrons		1 p, 1 e, 0 n		()	
Common ion (example of a compound)		H ¹⁺ ion (eg HCl)		()	

- 6 (a) List 10 different metals.
 (b) List five different non-metals.
 (c) State where other metals are placed on the periodic table. Where are the non-metals listed?
- 7 (a) The large area between the two ends of the table includes the transition metals. Name five of these.
 (b) Element number 39 was discovered in Ytterby in Sweden. What is its name?
 (c) Mercury is sometimes called quicksilver. Why?

What happened next?

Mendeléeve's table did not include any of the noble gases. The first of these was not discovered until 1894 by William Ramsey. The British scientist Henry Mosley (1887–1915) changed Mendeléeve's periodic law. When Mendeléeve wrote 'The properties of elements change periodically with their atomic mass', he had no knowledge of the protons or neutrons in the atom. Mosley changed the law to read 'The properties of elements change periodically with their atomic number'. This change allowed many more elements to fit neatly into the table.

Mendeléeve did not predict any of the artificial elements, as he had only predicted elements for the gaps in the table, not for the end of the table. These artificial elements have only been discovered since the discovery of atomic energy. Most of these artificial elements have atoms which are larger than atoms of uranium. So far these elements fit Mendeléeve's table. Many of the early transuranic (larger than uranium) metals were discovered by Glenn Seaborg and his team at the University of Chicago during World War II. They produced curium (element number 96) and americium (element number 95).

- 8 (a) Krypton is said to be a noble gas. Name some other elements in this family.
 (b) What is a proton?
 (c) What is a neutron?
- 9 (a) What is element number 101?
 (b) After whom was this element named?

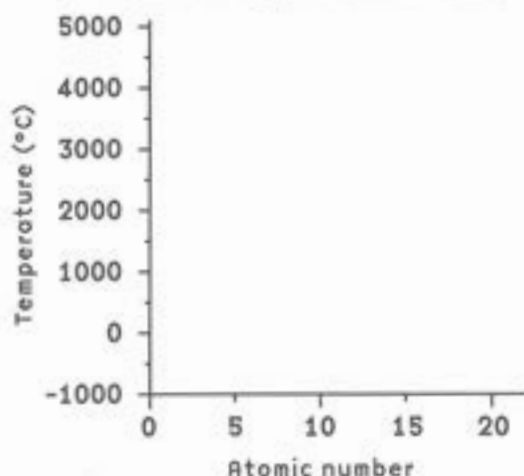
Melting and Boiling Points

The melting and boiling points of the elements can be used to describe the type of attractive forces that exist between the atoms. Elements found as single atoms or small molecules have very low melting and boiling points. Other elements that form complex crystals have the highest melting and boiling points. The melting and boiling points for the first 20 elements are given in the table.

- 1 Use the information in the table to graph the melting and boiling points on the grid.
 - (a) Describe the shape of the graph.
 - (b) Describe any pattern you can see.
- 2
 - (a) Which element has the highest boiling point? What is its boiling point?
 - (b) Which has the lowest melting point? What is its melting point?
 - (c) Describe the elements with negative melting and boiling points?
 - (d) Bromine (element number 35) is placed below chlorine in the table. Estimate its melting and boiling points.

At No	Element	Symbol	mp (°C)	bp (°C)
1	hydrogen	H	-259	-253
2	helium	He	-270	-269
3	lithium	Li	180	1330
4	beryllium	Be	1280	2480
5	boron	B	2030	3900
6	carbon	C	3730	4830
7	nitrogen	N	-210	-196
8	oxygen	O	-219	-183
9	fluorine	F	-220	-188
10	neon	Ne	-249	-246
11	sodium	Na	98	892
12	magnesium	Mg	650	1110
13	aluminium	Al	660	2450
14	silicon	Si	1410	2680
15	phosphorus	P	44	280
16	sulfur	S	119	445
17	chlorine	Cl	-101	-35
18	argon	Ar	-189	-186
19	potassium	K	64	760
20	calcium	Ca	838	1490

Melting and boiling points of Elements

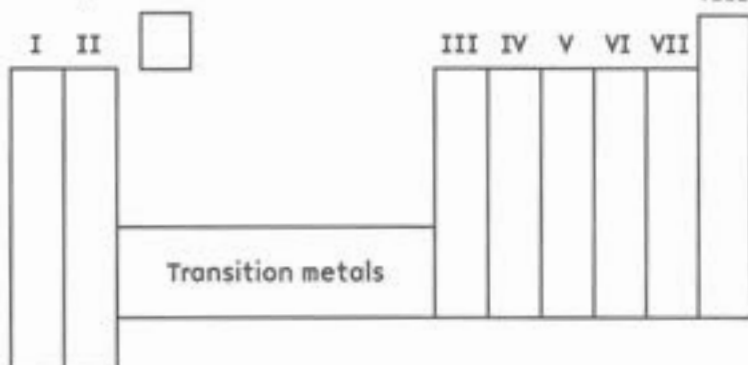


Groups

- 3 Name the first element in each of the different numbered groups.

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

Groups



Group I

The elements lithium (Li), sodium (Na), potassium (K), rubidium (Rb) and caesium (Cs) in the first group (vertical column) of the periodic table are all similar to look at and have similar reactions. All of these elements are soft solids. They all react very strongly with the halogen elements from group VII. Lithium, the smallest element in group I, is the least reactive and caesium is the most reactive.

- 4 (a) Hydrogen is often on its own. Label hydrogen in its correct space on the diagram on the previous page.
 (b) Label lithium and caesium in their correct places on the diagram.
 (c) Group VIII are called the noble gases. List some examples.

Group II

The second group of elements include beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr) and barium (Ba). The members of this group are all hard, shiny solids. Most members of this group react with halogens. Beryllium, the smallest in this group, is the least reactive and barium, the largest of the group, is the most reactive.

Group VII

Group VII elements are non-metals and the smaller elements are gases at room temperature. Fluorine (F), chlorine (Cl), bromine (Br) and iodine (I) are in this group. Fluorine, the smallest, is extremely reactive with hydrogen, other non-metals, water and metals. Iodine, the largest, reacts slowly with some metals and non-metals.

- 5 (a) Label beryllium, barium, fluorine and iodine in their correct places on the diagram.

LOOK IT UP 🔍

- (b) Describe any difference in reactivity between elements with the largest and smallest atoms in:
 Group I
 Group II
 Group III.

Mendelevium (Md)

Mendelevium was named after Mendelée. It was discovered by Albert Ghiorso in 1955 after firing the small helium nucleus at larger atoms. It has an atomic number of 101, making it a very large atom. This new element was named after Mendelée because he had organised the elements and predicted new ones.

- 6 (a) Other elements were named after famous scientists. Complete the table and describe in more detail the characteristics of one of these elements.

- (b) What important scientific work was done by
 i Enrico Fermi?
 ii Marie Curie?

Atomic Number	Name	Scientist
99		
	Lawrencium	
		Dmitri Mendelée
102		
		Marie Curie

- 7 (a) Who discovered mendelevium?

LOOK IT UP 🔍

- (b) How did this discovery come about?

GROUP WORK 🗨️

- 8 Why were other elements given the names they have? Some interesting ones are: americium, berkelium, californium, europium, francium, germanium, krypton, mercury, neptunium, plutonium, polonium, promethium, radium, samarium and technetium.

Isaac Newton was born on Christmas Day in 1642 in Woolsthorpe, Lincolnshire, England. In 1656, Newton left school but he was still very interested in mathematics. Newton started at Trinity College in Cambridge in June 1661. During one summer, the plague killed many people in London. Newton was sent to the country for protection and while there he invented a new method of mathematical calculation. Later it would be called calculus but Newton called it 'method of fluxion'. It was based on finding the area under a curve. Because of this remarkable piece of work, Newton was made Professor of Mathematics at Cambridge when he was only 27 years old.

- 1 Newton developed his knowledge through education. How could your education develop an understanding of science?
- 2 (a) What is something that you would like to study or discover?
(b) How would this discovery be useful?

The story so far

Air Pressure

The great English chemist Robert Boyle (1627–91) had done experiments on air. His research produced famous gas laws and gas equations. Simply stated, in a given amount of gas, if pressure is increased then volume is decreased (Boyle's Law). Boyle was very good at observation and experimental science and this scientific method would influence Newton and many other scientists who came later. The results of Boyle's work are also used in the technology of our modern world.



- 3 (a) Rewrite Boyle's Law in your own words, showing what might happen to the pressure if the volume of a container of gas is increased.

Another gas law can be summarised as

$$P_i V_i = P_f V_f$$

where P_i and V_i are the initial pressure and volume, and P_f and V_f are the final pressure and volume.

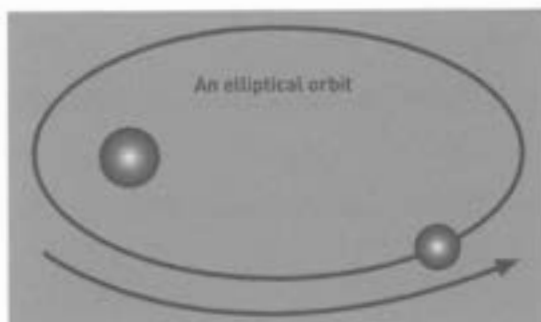
- (b) If 3 L [litres] of gas [V_i] under a pressure of 1000 kPa [kilopascals] [P_i] have the pressure increased to 2000 kPa [P_f], what is the new volume [V_f]?

GROUP WORK

- 4 (a) Work with your group to make a list of the instances where gas volume and pressure is important in our society.
(b) Choose one of these examples and describe how it affects your life.

Moving Planets

Johannes Kepler had the idea that the planets move in the shape of an ellipse (not a circle as many people believed), with the sun at one focus. This is now known as Kepler's First Law. An extension of this idea was that the force attracting the planets towards the sun would vary according to their distances from it. (It would be a larger force when the planet was close to the sun, and a smaller force when the planet was further away.)



- 5 (a) Draw an ellipse and a circle to show the difference.
- (b) The orbit of the Earth around the sun is an ellipse. What does this mean about the distance of the Earth from the sun?
- (c) Rewrite Kepler's First Law in your own words.

Kepler's law is summarised by the inverse square law; $F \propto \frac{1}{d^2}$ = Force is proportional to 1 divided by distance squared.

The symbol $\propto$ means 'is proportional to'.

- 6 If the distance [d] from the Earth to an orbiting satellite is doubled from 1 million km to 2 million km:
 - (a) What is the relative change in the value of the gravitational force [F]?
 - (b) How far away would the satellite be if the force was $\frac{1}{3}$ the original value?

What did Newton do?

At university Newton studied the ideas of many great thinkers, including the great English chemist Robert Boyle. Another famous scientist whose ideas he studied was Johannes Kepler.

Newton's first lectures as Professor of Mathematics dealt with optics, and included some discoveries he had made during the years of the plague. Newton concluded that white light is not as simple as it appears. When he passed a thin beam of sunlight through a glass prism Newton noticed a collection or spectrum of colours, beginning with red and ranging through orange, yellow, green, blue, indigo and violet, that formed on the wall and floor opposite. Even though Newton could not tell exactly how many colours there were, because they seemed to blend together, he felt there must be seven —this being the perfect number.

- 7 (a) Draw a diagram showing how the seven colours of the spectrum were produced by Newton.

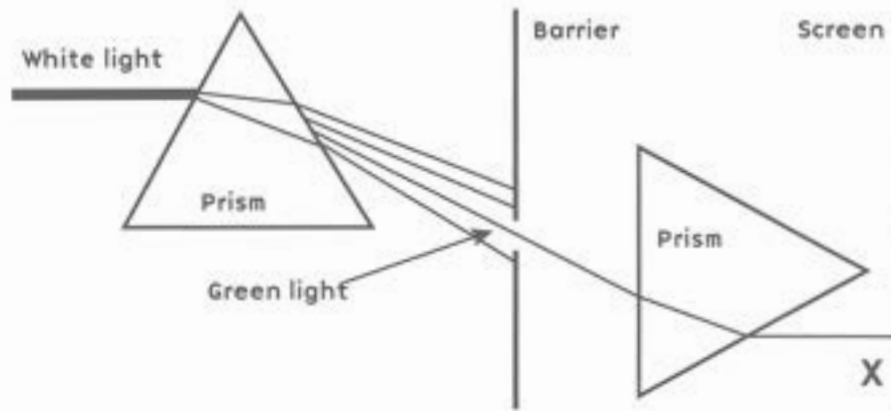
LOOK IT UP

- (b) Today we often only describe six colours (leaving out indigo). Why is that?
- (c) What is outside the red colour?
- (d) What is outside the violet colour?

What did Newton conclude about light?

Newton knew the phenomenon couldn't be due simply to refraction, or bending of light, because the different colours should all be refracted the same amount, and not spread out on the wall. Newton concluded that white light is really a mixture of many different types of light, and that these different rays of light are refracted at slightly different angles. Each different ray of light produces a different colour. Newton made this conclusion after carrying out the following experiment.

Newton passed one colour through a second prism and there was no change in this colour. The single colour was only bent by refraction as it passed through the second prism.



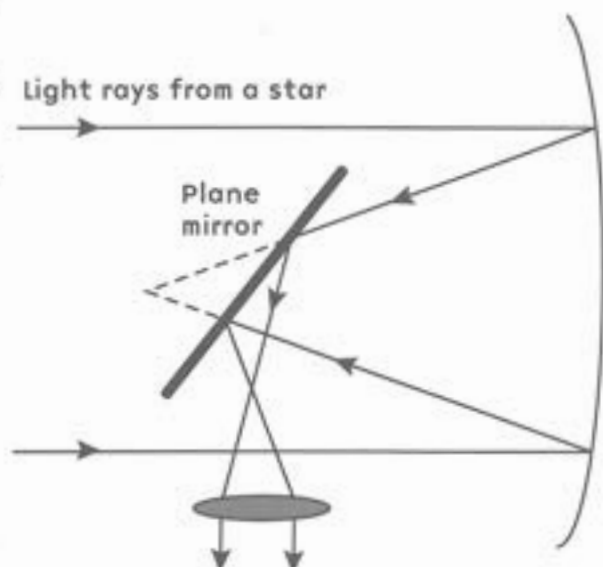
- 8 The process called dispersion is shown in the diagram. What colour or colours might you see at point X on the screen at the right of the diagram?

LOOK IT UP 🔍

- 9 (a) What is the difference between refraction and dispersion?
(b) Describe reflection and diffraction as they relate to light.

On the basis of these discoveries Newton invented a new type of telescope that made the images more clear. People are still using and improving on Newton's design today. A simple ray diagram of his telescope is shown below.

- 10 (a) What happens to the light at the concave mirror?
(b) What is the use of the plane mirror?
(c) At what angle should the plane mirror be placed? Why?
(d) What is the purpose of the convex lens?
(e) On the diagram, label:
i focal point
ii lens
iii eye position.



What did Newton conclude about motion and gravity?

Newton's greatest work was on mechanics and the theory of universal gravitation. He began this research in the plague years, and you can believe the story about the apple falling from a tree in his garden if you like!

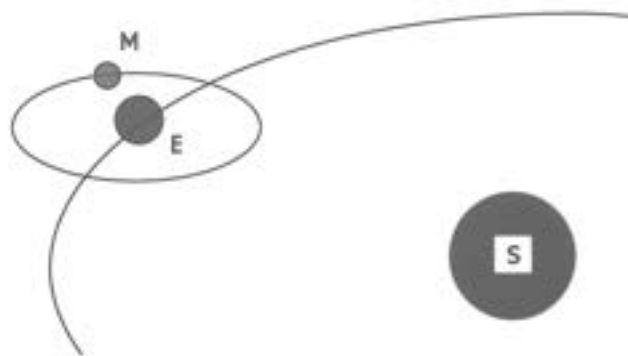
- 11 (a) Explain why the apple falls from the tree.
(b) Do objects always fall down? Explain.

Laws of Motion

Newton made up three Laws of Motion. **Law 1** says that if an object is not moving, or is moving at a constant velocity, then it will keep doing that until some force (push or pull) acts on it. **Law 2** relates the size of this force and the mass of the object to the resulting change in velocity or acceleration. The formula is written as $F = ma$ (Force = mass $\times$ acceleration). **Law 3** shows that for every force there is an equal and opposite force, which explains why we don't fall through the floor of the house. The floorboards push up on the bottom of our feet with the same force as gravity pulls us down. This means there is no net (ultimate or resulting) force, so we do not go up or down. Force is measured in newtons (N), named after Newton.

- 12 Rewrite Newton's three Laws of Motion in your own words, to indicate your understanding of what they mean.
- 13 (a) Using the relationship from the second law ($F = ma$), calculate the acceleration of a 1000 kg car with an engine pushing it with 500N of force.
(b) If the car slows down at a rate of 5 m.s^{-2} (metres per second per second), what is the force slowing it down?
(c) Astronauts in space can move from one place to another by small jet propulsion units. Using the Third Law, describe how these units cause the astronaut to move.

As well as the motion of objects in straight lines, Newton also studied circular motion. In 1666 he figured out that the gravity of the Earth extended out from the Earth's surface, reaching even to the Moon. Newton published this result in his book *Principia*. Newton's law of universal gravitation states that every body of mass attracts every other body of mass with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.



The sun (S) with the Earth (E) and the Earth's Moon (M) in orbit around it.

- 14 (a) Using the diagram, describe what gravitational forces might be acting on each of these three objects in space.
(b) How would these affect (or be affected by) the outer planets (such as Neptune and Pluto)?

What happened next?

The ideas of mass and gravity were continued to be discussed by many scientists including Albert Einstein who developed a whole theory of time, space and universal gravitation. Currently scientists are looking at the gravitational effects on light. The movements of planets around stars, as well as the movement of planets in solar systems other than our own, are studied with reference to Newton's work. Besides providing a foundation for researchers in astronomy, Newton's discoveries are the basis for understanding the motion of any moving object, whether it be a cricket ball or a racing car, and the forces which cause these objects to speed up or slow down.

Ernest Rutherford (1871–1937)

9a

Ernest Rutherford was labelled as English because he did most of his research in England, but he was actually born near Nelson on the South Island of New Zealand. Rutherford went to the Cavendish Laboratory at the University of Cambridge and worked under the leadership of Sir Joseph John (J. J.) Thomson (1856–1940). Thomson had received the Nobel Prize for discovering the electron. Rutherford's early work with Thomson led to investigations of electricity and radiation, and they later studied radioactivity.

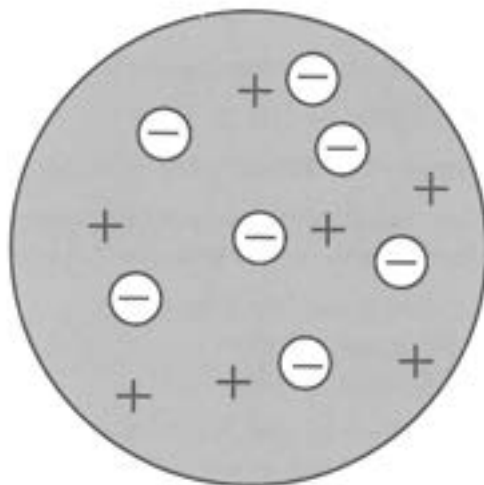
In 1898, Rutherford became Professor of Physics at McGill University, Canada. While there, he discovered several radioactive elements. In 1919 Rutherford moved back to Cambridge as Director of the Cavendish Laboratory. During this time his team discovered artificial disintegration of the atom, the neutron, and what happens as particles are bombarded by other accelerated particles. All this led to increased knowledge of the structure of the atom.

- In what places did Rutherford live and work?
 - What advantage do you think the Cavendish Laboratory had over universities in New Zealand or Canada?
- What did J. J. Thomson discover?

The story so far

The accepted structure of the atom at the time was called the 'plum pudding' model. This idea or model of the atom contained positive and negative charges mixed into a ball. It was called plum pudding because it reminded people of a cake with dried fruit through it.

- In the diagram of the atom shown, how many
(a) positive charges in the atom?
(b) negative charges?
(c) Are the charges in any pattern?
- What name would you use to describe this picture of an atom?
- If you had a microscopic bullet and fired it at this atom, what would you expect to happen?



What did Rutherford do?

Rutherford found two kinds of radiation and he called them alpha α and beta β rays (beta particles were shown later to be electrons). Rutherford also showed that the alpha rays were the same as the nucleus of helium atoms and were, in fact, particles. In 1909 Rutherford used these α particles to study the structure of matter. In his experiment, Rutherford fired α particles towards very, very thin sheets of gold. He expected most of the α particles to travel right through the thin sheets of metal, but instead some came back at him. Rutherford was amazed: 'It was almost as incredible, as if you fired a fifteen-inch shell at a piece of tissue paper and it came back and hit you'.

- What is an α particle?
 - What is another name for a β particle?
 - What did Rutherford mean when he said, 'as if you fired a fifteen-inch shell at a piece of tissue paper'?

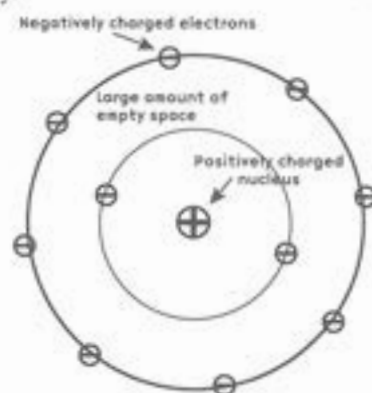
What did Rutherford conclude?

Rutherford's experiments produced a greater than expected deflection. This showed that there was a large dense particle inside the atom. Based on these results, Rutherford calculated that a massive nucleus was

concentrated in only 1/10 000th of the volume of the atom. The rest of the mass, and the negative charges (made of electrons), were spread out in a sphere of very low density.

- 7 (a) How many electrons in the diagram?
 (b) What is the charge of the nucleus?
 (c) What makes up most of the atom?

This model of the atom is the one we commonly see in science books, but the nucleus of any atom is so small compared to the amount of empty space between the nucleus and the electrons that it is very difficult to visualise or draw in a diagram. In fact, if the nucleus was the size of this dot (.) and you placed the dot in the centre of a sporting field such as the Sydney Cricket Ground, then the first row of electrons would be on the boundary fence.



- 8 The diagram is not exactly accurate. How would you change it?

Following on from Rutherford's research the neutron was discovered by James Chadwick and research began on different isotopes of elements. An isotope exists when an element has different numbers of neutrons in the nucleus. Rutherford's later work used the idea of atoms having protons and neutrons in the nucleus surrounded by electrons to study the process of transmutation. Transmutation is the changing of one element to another by nuclear changes or decay. This occurs if protons are emitted (or lost) from a nucleus or if a neutron changes into a proton.

- 9 (a) Who is credited with the discovery of the neutron?
 (b) What is a neutron?
- 10 (a) How can one element be changed to another?
 (b) What is this called?

One example of transmutation is the formation of gold (to answer the dream of the alchemists!):



Hydrogen 2 + Mercury 199 → α particle + Gold 197

In this reaction the isotope of hydrogen with 1 proton (bottom number) and 1 neutron (1 + 1 = 2, top number) is fired at an isotope of mercury (80 protons and 119 neutrons), producing a helium nucleus and gold.

- 11 What is described by the
 (a) lower number?
 (b) upper number?
- 12 (a) What is used for the bullet?
 (b) What is the formula for an α particle?
 (c) How many protons in any gold atom?
 (d) How many neutrons in this gold?

What happens next?

Only 90 of the elements have been found naturally on Earth. Of the first 92 elements, the two which are not found naturally are technetium and promethium. This is because they are unstable and almost immediately breakdown and change into more stable elements. This atomic breakdown releases radiation. As a result, these atoms are radioactive. Some elements can be found as different isotopes and some of these are also radioactive. Further research into radioisotopes, using accelerators such as those at the Fermi Laboratory, is being undertaken. Many of these radioisotopes are very useful.

LOOK IT UP 🔍

- 13 Describe any uses you know of for radioisotopes in medicine or industry.

GROUP WORK 🗨️

- 14 Write a letter to ANSTO at Lucas Heights asking for information on what radioisotopes they produce there, and what are some of the uses of these radioisotopes.

Radioisotopes are used for many different reasons. Below are 10 different uses, organised according to whether they are used for mainly medical or engineering purposes.

Biological or Medical

Uses	Reason	How does it work?
Radiology	X-rays are used to diagnose tooth decay, bone fractures and detection of some tumours.	X-rays are stopped by dense tissue and can be photographed to produce a density map of parts of the body, such as bones and teeth.

- 1
 - (a) Have you or a friend had an X-ray? If so, what part of the body was X-rayed?
 - (b) What did the radiograph look like?
 - (c) How did it help in the particular medical situation?
 - (d) X-rays are produced by radioactive decay and can cause problems in cells, particularly in unborn children. What protection is used by radiologists and others who use X-rays for medical purposes?

PET	Position Emission Tomography. Certain nuclides such as oxygen -15, nitrogen -13, and carbon -11, emit positrons.	Positrons are almost immediately destroyed when they meet an electron. This reaction releases γ radiation which is used to pinpoint the positrons of individual molecules in the body.
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- 2
 - (a) What do the terms oxygen - 15, nitrogen - 13 and carbon - 11 mean?
 - (b) A positron has the same mass as an electron but the opposite charge. What charge do positrons have?

Tracers	γ ray emitters are injected or ingested to trace pathways.	The emitters show their path through the system or are selected by the system to congregate at specific sites such as the brain, kidneys, or bones.
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- 3
 - (a) What is γ radiation?
 - (b) What does 'emitter' mean?
 - (c) What element or mineral is collected by bones?

Radiation treatment	Radiation destroys cancer cells.	Cancer cells are fast growing and are very sensitive to radiation. Surrounding healthy tissue is less sensitive and so there is a range of radiation which will kill some cells but not others.
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- 4
 - (a) Often we are affected by solar radiation (sunburn). What happens to our skin?
 - (b) How could radiation affect the growth or reproduction of a cell?
 - (c) How can radiation treatment affect cancer cells more than healthy tissue?

Industrial or Technological Uses

Uses	Reason	How does it work?
Industrial radiography	Density mapping using γ radiation.	Similar to biological radiography, a density map can show weakness in structures or joins and can also show areas where impurities are concentrated.

- 5 (a) How can impurities in a metal or other material be 'seen' by using γ rays?
 (b) What is a density map?

Neutron radiography	Neutron capture by small atoms.	Slow-moving neutrons will be captured by some small nuclei such as hydrogen, cadmium and boron. Molecules with these atoms (such as water) can be detected and can be used to detect corrosion and the presence of explosives in luggage.
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- 6 (a) What would be the advantage of finding water inside different materials?
 (b) How could explosives be 'seen' by neutron radiography?

Mineral analysis	γ rays can induce mineral fluorescence.	The amount of fluorescence is related to the concentration of the particular mineral, and changes in industrial processes, waste control or prospecting for mineral sources can be studied.
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- 7 (a) What is fluorescence?
 (b) How can this analysis help in waste control?

Gauging	Thickness mapping using γ radiation.	Using a source on one side and detectors on the other, monitoring of packaging and thickness of such diverse materials as tin plate, aluminium foil and plastic wrap can be studied.
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- 8 Explain how gauging with radiation can be used to keep plastic wrap a constant thickness as it is being manufactured?

Soil erosion	Using fallout from nuclear testing.	Caesium - 137 was deposited around the world due to the atomic tests of the 1950s. This can now be detected and traced to show the movement of topsoil due to erosion.
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- 9 (a) What is weathering?
 (b) What is erosion?
 (c) What happened to the surface of the Earth during the nuclear bomb testing?
 (d) How can we use this to test for erosion and weathering?

Fertiliser uptake	Phosphorus - 32 is taken up by plants.	Including the phosphorus isotope in fertiliser can give information about take up rates, and show ways of targeting fertiliser in terms of most useful uptake time or weather conditions.
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- 10 Explain how farmers can determine the best time to fertilise their crops using the radioactive isotope phosphorus - 32.

Flossie Wong-Staal's original name was Wong Yee Ching and she was born in China in 1947. In 1952 her family settled in Hong Kong. While at school, she was given the English name Flossie after Typhoon Flossie, which had struck Hong Kong the week before. Wong-Staal enjoyed studying science and maths and her family was very supportive of Wong-Staal continuing her education. In 1965, she emigrated to the United States to study at the University of California, where she specialised in molecular biology and continued as a postgraduate student in bacteriology.

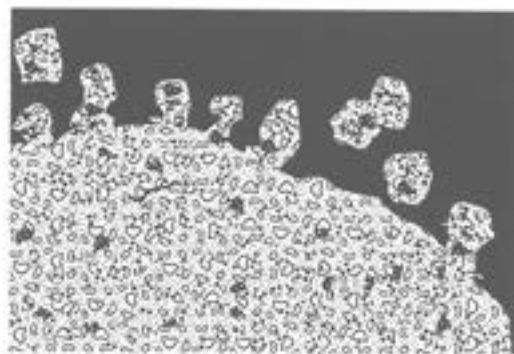
After completing her doctorate she took a position with the National Cancer Institute in Bethesda, Maryland, where she was the first researcher to copy the human immunodeficiency virus (HIV). The cloning of the virus that causes AIDS allowed Wong-Staal and other scientists to determine the structure of HIV. She has continued to work in AIDS research, specifically gene therapy.

- 1 (a) What is cancer?
(b) List one cancer and describe its effect.
- 2 (a) What does the term 'cloning' mean?
(b) Suggest reasons why it is important to know the structure of the virus?

The story so far

In the 1970s, a number of rare cancers and other serious infections were becoming more and more common in healthy people. Acquired Immune Deficiency Syndrome, or AIDS, was formally described in 1981 as a syndrome. It was at first seen as affecting homosexual men, people who used intravenous drugs, haemophiliacs, and some people who had been given blood transfusions. Today we know that specific behaviours such as unsafe sex, the sharing of needles, or infection from blood, increases the risk of getting AIDS.

- 3 (a) What is AIDS?
(b) What are regarded as high-risk activities in the spread of AIDS?



HIV Virus

AIDS is caused by infection with the human immunodeficiency virus (HIV). This virus attacks specially selected cells in the immune system of a person so that the system does not work correctly. Normally the body's immune system resists harmful organisms. Being infected with HIV means that the immune system cannot work as it normally does. This might not be noticed, sometimes for many years. In most cases, however, the AIDS virus means that the body allows invasion by various infectious diseases and unusual cancers. In some cases the virus even attacks brain cells. This can lead to psychological problems.

In the early 1980s the AIDS epidemic was first recognised and researchers worked at many research laboratories around the world. The job of the researchers was made more difficult due to AIDS being defined as a syndrome because it shows symptoms of many different illnesses. The National Cancer Institute in which Wong-Staal was working with Robert Gallo was a major research centre in the fight against AIDS.

- 4 (a) What causes AIDS?
(b) How does this virus work?
(c) Why was AIDS more difficult to isolate than other diseases?

What did Wong-Staal do?

At the Cancer Institute, they used a special group of viruses called retroviruses. The human retroviruses (human immunodeficiency virus and human T-cell leukaemia virus) were both used because of the way they react with the cells they infect. They were also used to study the development of vaccines.

- 5 What is a retrovirus?

What did Wong-Staal conclude?

In 1983 Wong-Staal and her colleagues discovered HIV at the same time as a French researcher. This caused a lot of national rivalry back and forth across the Atlantic Ocean. As Wong-Staal was personally responsible for the first cloning and describing of HIV, her research started the development of tests for HIV. These tests are used to screen blood donations and to test people for the presence of the virus in their bodies.

GROUP WORK

- 6 [a] What are the advantages of testing for HIV?
[b] Why is it important to screen blood donations?

What happens next?

At the moment, no cure or vaccine exists for AIDS. A large number of the people infected with HIV may not even know that they carry the virus or that they are capable of spreading it. HIV infection and AIDS represent one of the most difficult health problems in today's world.

Fighting the effect of HIV is a major challenge for scientists and medical researchers. Stopping the spread of AIDS is a major problem for health-care workers and public health.

Wong-Staal's current focus is to discover a vaccine for the AIDS virus, perhaps using a made-up (synthetic) copy of HIV. She is involved with biotechnology companies developing drugs that might destroy the reproductive cycle of the virus. Wong-Staal is also working to treat those already suffering from the disease.

- 7 [a] What is the next task that Flossie Wong-Staal has set herself?
[b] How does she hope to achieve that result?

Because there is not yet a vaccine which is effective against AIDS, education leading to risk reduction remains the most important aspect of the fight against this syndrome. The further spread of AIDS can be stopped by avoiding high-risk behaviour. Education is vital, and in our society it is easy to present the message about safe behaviours through television. Unfortunately, even though the number of new AIDS cases is dropping, there are still many people who think that it won't happen to them and don't pay attention to the advice.

In many other countries that don't have mass education campaigns, the proportion of AIDS sufferers in the population is very high and growing. The other problems in many of these societies, such as war or famine, reduces the effect of the education campaigns that are available. Success in the war against AIDS still has a long way to go.

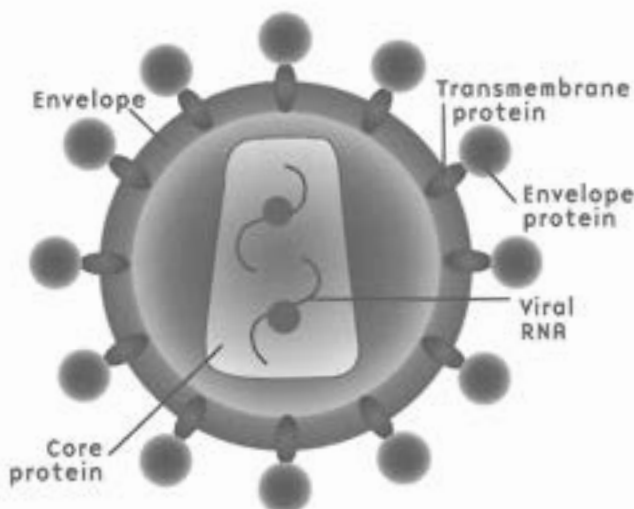
GROUP WORK

- 8 Discuss with your group the 'high-risk' behaviours. Imagine your job is to develop an education campaign for your suburb or town to warn against high-risk behaviour.
- [a] What particular behaviour will your campaign target?
[b] Why did you choose this particular behaviour?
[c] How does this behaviour lead to the spread of AIDS?
[d] On your own paper present your poster or advertising concept.

HIV is the virus most people believe causes AIDS. However, some scientists believe that HIV can only cause AIDS while acting in the presence of another virus, which has not yet been identified. Because most researchers believe that HIV is either the only or main cause of AIDS, HIV is often referred to as the 'AIDS virus'. AIDS spreads from one individual to another in body fluids such as blood or semen. It is most often spread through blood contact, unsafe sex practices or unclean surgical instruments or needles.

LOOK IT UP 🔍

- 1 (a) What is a virus?
(b) How do viral infections differ from bacterial and fungal infections?
- 2 (a) Using the letters below, give meanings for the words which form the acronym AIDS.
A Acquired
I Immune
D Deficiency
S Syndrome
(b) Using the letters below, give meanings for the words which form the acronym HIV.
H Human
I Immunodeficiency
V Virus



The internal structure of HIV

The current AIDS definition includes the following lung diseases: pneumocystis carinii pneumonia (PCP), and pulmonary tuberculosis. Cancers include Kaposi's sarcoma (KS), non-Hodgkin's lymphoma, primary lymphoma of the brain, and invasive cervical cancer. There are many more. These diseases are part of the definition of symptoms that an AIDS sufferer could present. If a person has more than one of these diseases, it is highly likely they have AIDS.

HIV continuum

Generally if you feel sick, you are sick; if you feel healthy, you are healthy. However, someone may have HIV and still not feel ill. The symptoms of illness may not be noticed until a long time after infection, if at all. The sequence of steps in HIV infection is called a continuum. The continuum described is very similar for many people with HIV, but the time taken to reach each stage varies from person to person. For most people the progress of HIV disease takes several years, from infection to the development of severe immunodeficiency.

- 3 (a) What does the term 'continuum' mean when describing AIDS?
(b) How does the immune response protect most people from disease?
- 4 (a) Why do the symptoms take time to develop after infection?
(b) Why do you think people who get AIDS react in different ways to the same infection?

Infection

HIV enters the bloodstream and invades the cells. People with HIV are considered to be infectious immediately and continue to be infectious at all times. People may look healthy for many years despite the fact that they have HIV in their bodies. The only way to find out if someone is infected is by taking an HIV antibody test. People with acute HIV infection usually will not test HIV antibody positive immediately, because it takes the body approximately one to three months to produce antibodies against HIV.

- 5 [a] Are all people with AIDS infectious?
- [b] How can a person find out if they have AIDS?
- [c] Can the test be used immediately after someone has been infected? Explain.

Immune system decline

The virus appears to slowly damage the immune system for a number of years after infection. This damage can be seen in blood tests, even before actual symptoms are experienced. It is extremely important that people with HIV see a doctor even if they feel fine, because the virus could already be damaging their immune systems. Treating infected persons before symptoms have appeared has extended the lifespan of HIV-infected people.

- 6 [a] What does the body's immune system do?
- [b] Why is it important to begin treatment before symptoms appear?

Symptoms

There are no set symptoms for HIV infection or AIDS. Most people who become infected with HIV don't notice any immediate change, although some may suffer from a flu-like illness shortly after infection.

Once the immune system is attacked, the person may catch 'opportunistic infections' which are infections that are around us all the time and can normally be fought off by healthy people (the most common of these infections is pneumonia). According to recent research, some tumours or cancers are opportunistic infections. These can cause damage to the brain and nervous system, and result in symptoms which are not caused by HIV but by the opportunistic infections.

GROUP WORK

- 7 Why do you think that the initial symptom of AIDS in many people is a flu-like illness?
- 8 [a] What is an opportunistic infection?
- [b] Do opportunistic infections attack everybody?
- [c] Why doesn't everyone get the opportunistic infections?

HIV Testing

A blood test has been developed which can tell if someone has been infected with HIV, even if they feel well. When germs invade the body it produces special chemicals called antibodies to fight them. Different antibodies are made for each germ. If someone has antibodies to fight HIV in their blood we know that they must have HIV. The blood test looks for these antibodies.

GROUP WORK

- 9 [a] What situations would lead people to get themselves tested for HIV?
- [b] What change to their activities could have prevented this infection?

Suggested Answers

Nick Jacques

- Study of elements and their reactions.
 - Study of compounds found in living things.
- Structure means how something is made and function means how it works.
 - How structure affects what the bacteria does.
- Study microbes in the mouth.
- Mothers and close family members.
- Dental plaque.
 - Tooth and gum disease.
- Diseases of the mouth cause illness and use financial resources.
- Bad bacteria attach themselves to teeth.
- Look for chemicals produced by bacteria and evidence of decay.
 - Answers will vary but could include looking at what biofilms are produced after diets high in sugar, salt, fat, vitamins or minerals.
- Give bacteria instructions for colonisation.
 - In order to accurately study the instructions.
- The matrix made by the bacteria.
 - Complex sugar.
 - Sugar in our diet.
- To protect [our teeth].
 - Bacteria have different properties and do different things.
 - Biofilm bacteria are more difficult to kill and more likely to cause disease.
- Answers will vary.

Bacteria Extension

- One.
 - 3 billion years ago.
 - Studies bacteria and other micro-organisms.
- Produce cheese and yoghurt.
 - Boils, meningitis, black plague, tonsillitis.
- 10^9 (1 billion)
 - 10^7 (10 million)
 - 10^6 (1 million)
- They have a strong cell wall coating.
- Protozoa are animals, algae are plants with chlorophyll, and yeast has no chlorophyll.
 - They are all microscopic organisms.
- Whip-like structure used for movement.
- To use oxygen.
 - Without oxygen.
- 2, 4, 8, 16, 32, 64, 128.
 - i 8, ii 64, iii 512.
- Answers will vary but will include cleanliness, correct storage and protection.
- (a + b) Answers will vary.

Robert Koch

- Study how different peoples live.

- Study ancient civilisations.
 - Study bacteria and how they affect us.
- Answers could include diphtheria and plague.
 - Heart disease, accidents, stress.
 - Should include safety improvements.
 - Answers will vary.
- Microscope.
 - Stage.
 - i 10x.
 - ii 10x & 20x.
 - Focal knobs.
- So it can be accurately treated.
 - Answers will vary.
- The new colony has the same colour, shape and reaction to different stains.
- By allowing the immune system to fight against a copy or a harmless version of the disease.
 - Should include measles and hepatitis.
 - Injections to keep the body in a state of readiness.
- Answers will vary.

Koch's Postulates Extension

- Answers will vary.
- Organisms visible only with a microscope.
 - Single-celled living thing.
 - Living material with cell walls but no chlorophyll.
 - Growth of tiny fungi.
 - Change in a situation due to unwanted material.
 - Measures we take to protect our wellbeing.
- Answers will vary.
 - Should include washing and air protection.
 - Answers will vary but should include a description of isolating the growing bread mould (in a plastic container or within plastic wrap).
- Place mould on nutrient; keep in a warm place.
- Micro-organisms in water, air, soil and food.
 - Maintain immune system by keeping healthy.
 - Answers will vary.
- To remove micro-organisms from our hands.
 - Pets contain microbes in mouths and on fur.
- (a + b) Answers will vary and will include refrigeration and covering of food, and clean hands.
- Include refrigeration and clean hands.
 - Eating live bacteria in the meat.

Jocelyn Ann Laurence

- Answers will vary.
- Burning, cancer.
- (a, b + c) Most students are exposed to various emr.

- Exercise and working on hot days.
 - Headaches, thirst.
- Answers will vary.
- 2450 MHz and 864 MHz.
 - Phone service frequencies.
- Faster reproduction and growth.
- Show different chemicals in the cells.
 - To see which compounds are present.
- Magnification and resolving power.
 - Objects have to be in a vacuum.
- Show what might happen to cells due to radiation.
 - Meeting of scientists to discuss ideas and find out what other people are doing in their field of interest.
 - They can learn from others' ideas.
- Answers will vary.

Electron Microscope Extension

- Light and electrons both focused by lenses.
 - Electrons are focused by magnetic fields.
- Transmission electron microscope.
 - Bounce off the [relatively] large object.
 - Electrons go through (TEM), bounce off (SEM).
- To focus the electron beam.
- We 'see' the image at the image plane.
 - To focus the electrons in a tight beam.
- Computers, motors and stereos.
- Convex lenses focus to a point.
- Means how close two objects can be and still be seen as two objects [rather than one].
 - Objects can be 10 000 times smaller and still be distinguished.
- Answers will vary.
 - Advantages — being able to see over a range; disadvantages — blurring.
 - Advantages — size; disadvantages — ease of loss, cleaning.

Antoine Laurent Lavoisier

- Safety and convenience.
 - Divisible by ten and a set list of prefixes.
 - Lavoisier was viewed as supporting the taxation of the King.
- Crushing, boiling in water, then distillation.
 - i HCl, ii H_2SO_4 , iii HNO_3 , iv As v Zn, vi CH_3CH_2OH
- Releases smoke, CO_2 and water.
 - Releases phosphorus oxide.
 - Releases CO_2 and water.
 - Releases smoke, CO_2 and water.
 - Releases smoke, CO_2 and water.
- Unburnt or partly burnt particles of fuel.
- Oxygen and nitrogen.

- (b) Respiration.
- 6 (a) HgO has oxygen and mercury.
(b) 44g.
- 8 (a) Slow burning of glucose with oxygen in the cell to release energy for the cell.
(b) Glucose and oxygen.
(c) Carbon dioxide.
- 9 (a) Material where all the atoms are identical.
(b) Oxygen and nitrogen.
(c) Two or more elements in fixed proportions.
(d) Carbon dioxide and mercury oxide.
(e) Carbon and oxygen and mercury.
- 10 (a) Description of attributes and what occurs.
(b) Based on numerical changes.
(c) Weighs reactants and products and counts atoms.

Mixtures and Compounds Extension

- 1 (a) Drinking water and soil.
(b) They were the only items available.
(c) Trace elements give a different flavour.
- 2 (a) Air, soup.
(b) Ammonia, rust.
(c) Ions.
(d) Micro-organisms in the water.
- 3 (a) Salts, nitrogen.
- 4 (a) Sodium chloride.
(b) Sodium hydrogen carbonate.
(c) Copper sulfate.
(d) Sulfuric acid.
(e) Calcium hydroxide.
- 5 (a) Light bulb glows.
(b) Nothing.
(c) Goes black.
(d) Black deposit.
- 6 (a) Energy lights the bulb.
(b) Sulfate ions are losing electrons, forming a gas.
(c) Copper ions gaining electrons to form copper.
(d) Particles of copper fall to the bottom.
- 7 (a) Decomposed it.
(b) Breaks into elements.

Charles Lyell

- 1 (a) Yes.
(b-d) Answers will vary.
- 2 (a) Earth's history was a short period of violent geological activity.
(b) Earth's history was long and changes occurred much more slowly.
(c) Calculated the age of the Earth using the Bible.
(d) About 5600 years.
- 3 (a) Geological activities were the same as 'what was happening now'.
(b) The action of rivers and volcanic activity.
(c) Erosion is breaking up rocks and deposition is the production of sediments to form new rocks.
- 4 (a, b + c) Rock that has been changed by heat or pressure which destroys fossils.
- 5 (a) Stratigraphy describes the layers in which rocks are found.
Palaeontology is the study of

- fossils.
- (b) Early rocks formed are on the bottom and 'younger' rocks are on the top.
- 6 Measure the amount of CO₂ in the shells.
- 7 (a) Theory of evolution.
(b) Lyell supported idea of regular, slow change.

Changes in the Earth Extension

- 1 Earth formed from dust around the sun.
- 2 (a) Cooling of the Earth.
(c) Very little hard material in the body.
(d) Pilbara, Australia.
- 3 (b) Using CO₂.
(c) Relating to the sea.
- 4 (a) Size.
(b) Dinosaurs were once living organisms.
(c) Flowering plants — flowers, coated seed; Pine trees — naked seeds, no flowers.
(d) Sections of the Earth's crust move.
- 5 (a) Mountain building.
(b) Answers will vary but could include great catastrophe.
(c) Answers will vary.
- 6 Answers will vary.

Barbara McClintock

- 1 (a) Study of inheritance.
(b) Study of plants.
(c) Cell structure of plant inheritance.
- 2 (a) Medical researchers.
(b) Answers will vary.
- 3 (a + b) Monk who first studied inheritance.
(c) Particular feature of an organism.
(d + e) Answers will vary.
- 4 (a) To grow a healthy crop.
(b) Develop strains of wheat which grew well.
- 5 (a) Supported Mendel's results.
(b) Tells which gene is on which chromosome.
(c) Chromosomes in the nucleus.
- 6 (a) Understand its requirements.
(b) No — different organisms.
- 7 (a) Between chromosomes in the development of gametes to increase variation.
(b) Change.
(c) Radiation, chemicals.
- 8 Change does not survive in the environment.
- 9 (a) Part of the chromosome which moves.
(b) Basic structure of life.
(c) Jumping gene.
- 10 (a) Change in genes.
(b) Diversity — variation between individuals; enzyme — chemical produced by genetic control; population — number of individuals in a group.

Cells Extension

- 1 Observed and described cells.
- 2 All life is made of cells; all cells come from other cells; cells are contained inside a cell membrane; all are similar

in their chemical makeup and size.

- 3 (a) Mitochondria, ribosome, membrane, nucleus.
(b) Energy and nutrients.
- 4 (a) Control movement of molecules.
(b) Maintain shape.
(c) Flexibility.
(d) Answers will vary.
- 6 (a) Instructions in the genes.
(b) DNA.
(c) DNA sends information to ribosomes to make protein (enzymes).
- 7 (a) On endoplasmic reticulum.
(b) Make protein.
(c) Yes; for respiration.
- 8 (a) No; used for photosynthesis.
(b) Green pigment used for photosynthesis.
- 9 (a) Chromosomes double and cells split.
(b) Crossing over or swapping of DNA material.

Dmitri Ivanovich Mendeléev

- 1 (a) At least 103.
(b) Any larger than number 92 + technetium.
(c) Carbon, oxygen, gold, lead, tin, iron.
- 2 (a) I.
(b) Rb, Cs, Fr.
(c) One electron in the outer shell, very reactive metal.
- 3 (a) VII.
(b) I, As.
(c) Gas, very reactive, seven electrons in the outer shell.
- 4 (a) Be, Ca, Sr, Ba, Ra.
(b) Gold.
(c) I S, II U.
(d) Ga.
(e) Ge.
- 5 Li 3; Lithium; 1818; 3 p, 3 e, 4 n; Li¹⁺ ion; [eg LiCl].
- 6 (a) Na, K, Rb, Cs, Sr, Ba, Cr, Mn, Fe, Co.
(b) C, N, O, F, Ar, Kr, Xe.
(c) Metals on the left, non-metals on the right.
- 7 (a) Nb, Mo, Tc, Ru, Rh.
(b) Yttrium.
(c) Silver-coloured liquid.
- 8 (a) He, Ne, Kr, Xe, Rn.
(b) Positive-charged particle found in the nucleus.
(c) Neutral-charged particle found in the nucleus.
- 9 (a) Md.
(b) Mendeléev.

Periodic Table Extension

- 1 (a) Has regular peaks and troughs.
(b) The high and low points should be eight elements apart.
- 2 (a) Carbon at 4830 °C.
(b) Helium at -270 °C.
(c) Gas at room temperature.
(d) Melting point just higher than -101 °C, boiling point just higher than -35 °C.
- 3 Li; Be; B; C; N; O; F; Ne.
- 4 (c) Ne, Ar, Kr, Xe, Rn.
- 5 (b) The largest elements are more

reactive than the smallest because the extra electron is further away from the nucleus and therefore easier to lose.

- 6 (a) 99, einsteinium, Einstein; 100 fermium, Fermi; 103, lawrencium, Lawrence; 101, mendelevium, Mendeliev; 102, nobelium, Nobel; 96, curium, Marie Curie.
(b) i Manufacture of the first nuclear reactor.
ii Discovery of radium and work with radiation.
- 7 (a) Glenn Seaborg in 1955.
(b) Bombarded einsteinium with helium nuclei.
- 8 Answers will vary.

Isaac Newton

- 1 Answers will vary.
- 2 (a + b) Answers will vary.
- 3 (a) Volume up = pressure down.
(b) 1.5 L.
- 4 (a) Include tyres, aeroplanes, balloons.
(b) Answers will vary.
- 5 (b) Distance changes.
(c) Answers will vary.
- 6 (a) 25%.
(b) 1.73 times.
- 7 (b) Newton described seven because that was a special number but today we divide the spectrum into six bands of colour.
(c) Infra-red.
(d) Ultra-violet.
- 8 Yellow or green.
- 9 (a) Refraction is bending and dispersion is breaking into components.
(b) Reflection is bouncing off; diffraction is spreading out after passing through a small gap or hole.
- 10 (a) Reflects to a focal point.
(b) Changes the position of the focal point.
(c) 45° so new rays are perpendicular to the barrel.
(d) Return the light rays to parallel.
- 11 (a) Gravity pulls apple and Earth toward each other.
(b) Yes, by definition.
- 12 Answers will vary.
- 13 (a) 0.5 ms^{-2} .
(b) 5000 N.
(c) The force of the propulsion unit sends particles into space and the astronaut in the opposite direction.
- 14 (a) Each object is attracted by gravity to the other two. The size of the force is determined by the distance.
(b) Much less because they are much further away.

Ernest Rutherford

- 1 (a) New Zealand, England and Canada.
(b) It attracted scientists with different ideas.
- 2 Electron.
- 3 (a) 7.
(b) 7.
(c) No.
- 4 Answers will vary.

- 5 Answers will vary but should imply hit and bounce or hit and be absorbed.
- 6 (a) Helium nucleus.
(b) Electron.
(c) Very large bullet at a very thin target.
- 7 (a) 10.
(b) +10.
(c) Empty space.
- 8 Make the nucleus and electrons much smaller and further apart.
- 9 (a) James Chadwick.
(b) Neutral particle in the nucleus.
- 10 (a) Changing the number of protons.
(b) Transmutation.
- 11 (a) Number of protons.
(b) Number of protons plus neutrons (nucleons).
- 12 (a) Hydrogen ion.
(b) $\frac{1}{2}\text{He}$.
(c) 79.
(d) $197 - 79 = 118$.
- 13 Medical and engineering uses (scanning).

Radioisotopes Extension

- 1 (a, b + c) Answers will vary.
(d) Lead shielding.
- 2 (a) Different isotopes of these elements.
(b) Positive.
- 3 (a) Electromagnetic radiation like X-rays.
(b) Producer.
(c) Calcium.
- 4 (a) Cells burn.
(b) Radiation could destroy the cell or the instructions that control accurate reproduction.
(c) Cancer cells reproduce at a faster rate than ordinary tissue.
- 5 (a) Mapping different densities of soil or material.
(b) A density map is a diagram of the different densities of material in an area.
- 6 (a) Corrosion or rusting in iron materials is caused by water. Water also indicates leakage.
(b) They contain different compounds from the surrounding material.
- 7 (a) Light.
(b) Different materials produce different amounts of light and light differences can show its presence, absence or concentration.
- 8 By putting the thickness through a computer, the result of the test can determine whether the film is thinned or thickened during manufacture.
- 9 (a) Breakdown of rock particles.
(b) Removal of these broken particles to a new site.
(c) Radioactive materials were deposited in the topsoil from fallout caused by the explosions.
(d) Test to see if it remains in the soil, and if it occurs in new locations.
- 10 Checking the movement of the isotope can show whether it is used by the plant or remains in the soil.

Flossie Wong-Staal

- 1 (a) Fast growing cells which kill normal cells.
(b) Answers will vary but include lung cancer.
- 2 (a) Identical (living things growing from a single cell).
(b) Researching how to fight it needs knowledge of how a virus works.
- 3 (a) Acquired Immune Deficiency Syndrome.
(b) Unsafe sex, intravenous drug use.
- 4 (a) HIV virus.
(b) It destroys the human immune system.
(c) It was small and seemed to be caused by many different processes.
- 5 Very small disease-causing virus.
- 6 (a) Allows treatment to begin.
(b) Because the HIV virus can be passed to recipients of blood donations.
- 7 (a) To find a cure for AIDS.
(b) By copying HIV; developing drugs to destroy the reproductive cycle of virus.
- 8 (a, b, c + d) Answers will vary.

AIDS Extension

- 1 (a) Small living organism which can only exist in other cells.
(b) Viruses cannot be destroyed by antibiotics.
- 2 (a) A Acquired — obtained somewhere else;
I Immune — related to the immune system (protective);
D Deficiency — lack;
S Syndrome — general group of symptoms.
(b) H Human — people;
I Immunodeficiency — related to lack of immunity;
V Virus — small living cell.
- 3 (a) Big range of symptoms and stages.
(b) Immunity in the body to fight disease.
- 4 (a) Each body is an individual and different.
(b) Each body is an individual.
- 5 (a) No, but you cannot tell so precautions are always necessary.
(b) Blood test.
(c) No because time is needed for the chemicals and changed cells to develop.
- 6 (a) Protects the body from infection.
(b) The body starts to be destroyed before symptoms appear.
- 7 Those symptoms are often the first to appear when the body is infected.
- 8 (a) An infection which attacks when an immune system is under stress.
(b) Yes.
(c) Some have better immune systems than others.
- 10 (a) Active sex life with multiple partners; intravenous drug user; stepping on a discarded needle.
(b) Not sharing needles, using safe sex practices or best practice safety precautions.

Topic	Volume/page	Topic	Volume/page	Topic	Volume/page
AIDS	II/41, II/43	Franklin	I/37	oxygen	I/15
α particle	I/33, II/37	Galileo	I/41	Paleozoic Era	II/23
Archimedes	I/8	γ rays	I/33, II/39	parasites	I/27
Aristotle	I/5	gas	I/15	periodic table	II/31, II/29
atoll	I/29	genetics	II/25, II/27	PET	II/39
atom	II/37	geological history	I/29, II/21, II/23	photography	I/9
atomic number	II/30	gravity	II/36	photosynthesis	I/39
Avery	I/37	groups (elements)	II/31	planets	I/43
bacteria	II/5, II/7	helium	I/16	plastic	I/11
Baekeland	I/9	HIV	II/41, II/43	Plato	I/41
bakelite	I/10, I/11	Hooke	II/27	Pluto	I/44
Becquerel	I/33	Hutton	II/21	polymer	I/11
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